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1 – OVERVIEW

This procedure will attempt to assist the FE with isolating failures within the System Support Module (SSM). This procedure is not intended as a guide to troubleshooting the various problem scenarios associated with all the components driven or sourced by the SSM. The primary goal of this procedure is to assist the FE with determining if a system problem can be isolated to the SSM. If the problem can be isolated to the SSM, then the secondary goal is to help the FE isolate the problem to an SSM sub-component that can be replaced. A flowchart is provided that the FE will follow during the course of the troubleshooting process. The flowchart will, at certain points, refer the FE to various troubleshooting sections in this document. The FE will perform the instructions in the sections to determine if the problem is internal to the SSM. If the problem is determined to be internal to the SSM then the instructions will provide the FE with a troubleshooting process for isolating the problem down to a replaceable part.

Note: The SSM is not present in EXCITE3 forward production systems. It is present in systems upgraded from EXCITE to EXCITE3, but it's only responsibilities are to monitor RF power, provide power for the SRI3 and PAC, and provide all patient handling functions.

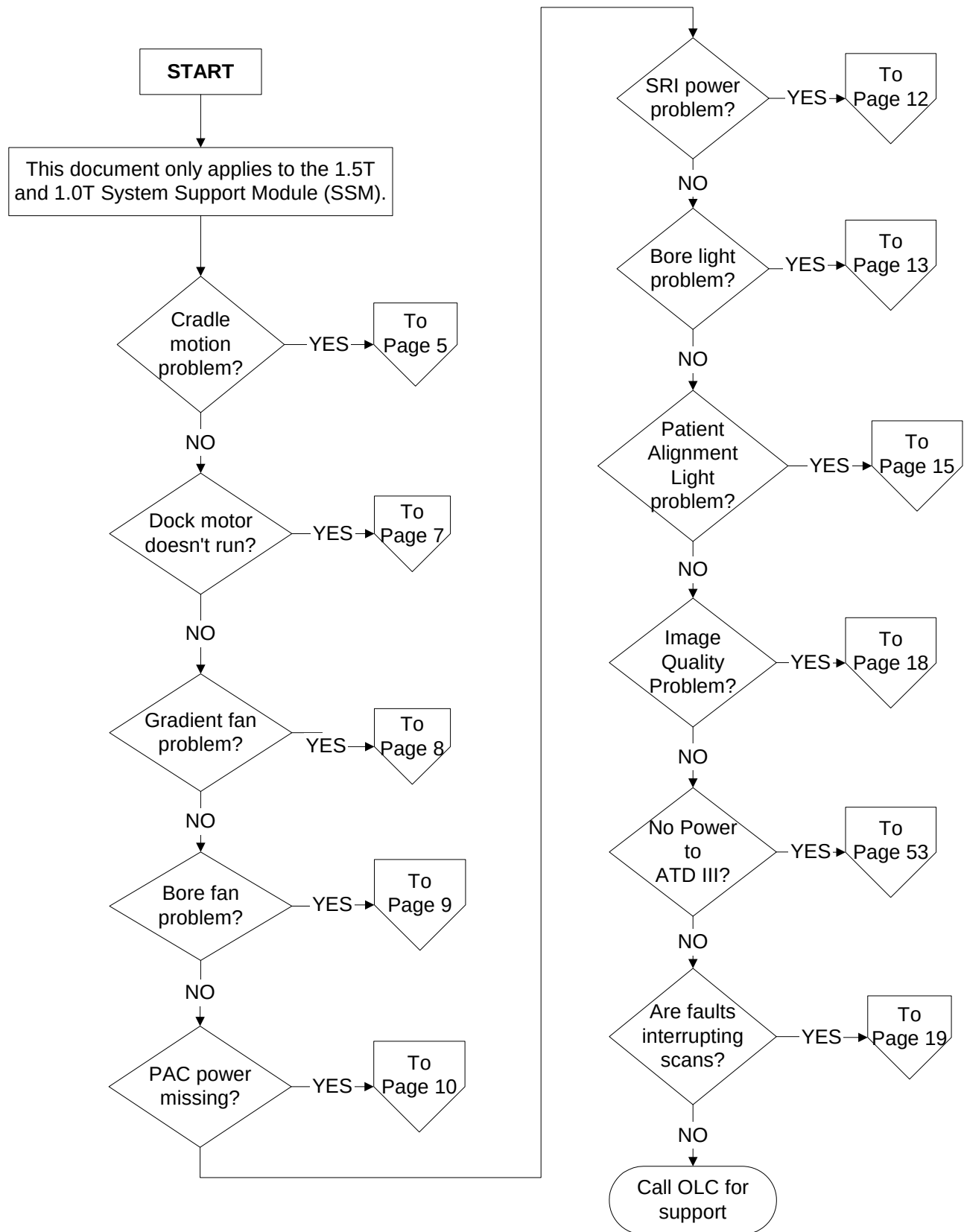
1-1 Required Tools

TABLE 1-1
EQUIPMENT REQUIRED

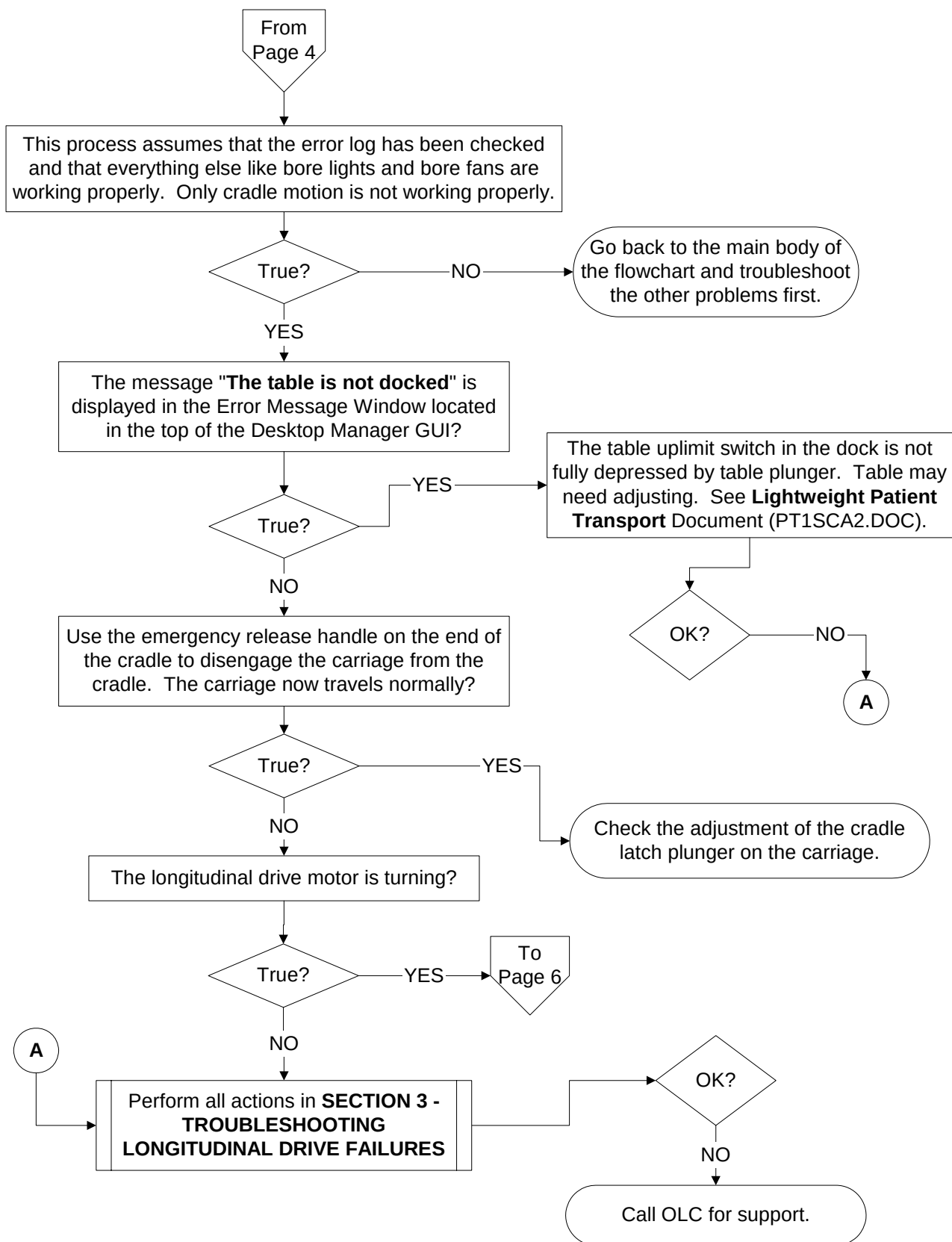
Item	Description	Part Number
1.	100 MHz Scope (equivalent or greater)	46-183029P61
2.	RF Power Measurement Kit	46-317724G1 or G2
3.	50 ohm, 200 Watt, 30dB Attenuator (dummy load) NOTE: Only required with above 46-317724G1 kit.	46-317724P14
4.	RF Test Cables Kit	46-255816G1
5.	Digital Multimeter (DMM)	46-194427P49
6.	TPS RF Service Interface Kit	46-301927G1
7.	Copper Tape (2 inch width)	46-258218P5

2 – 1.5T AND 1.0T SSM TROUBLESHOOTING FLOWCHART

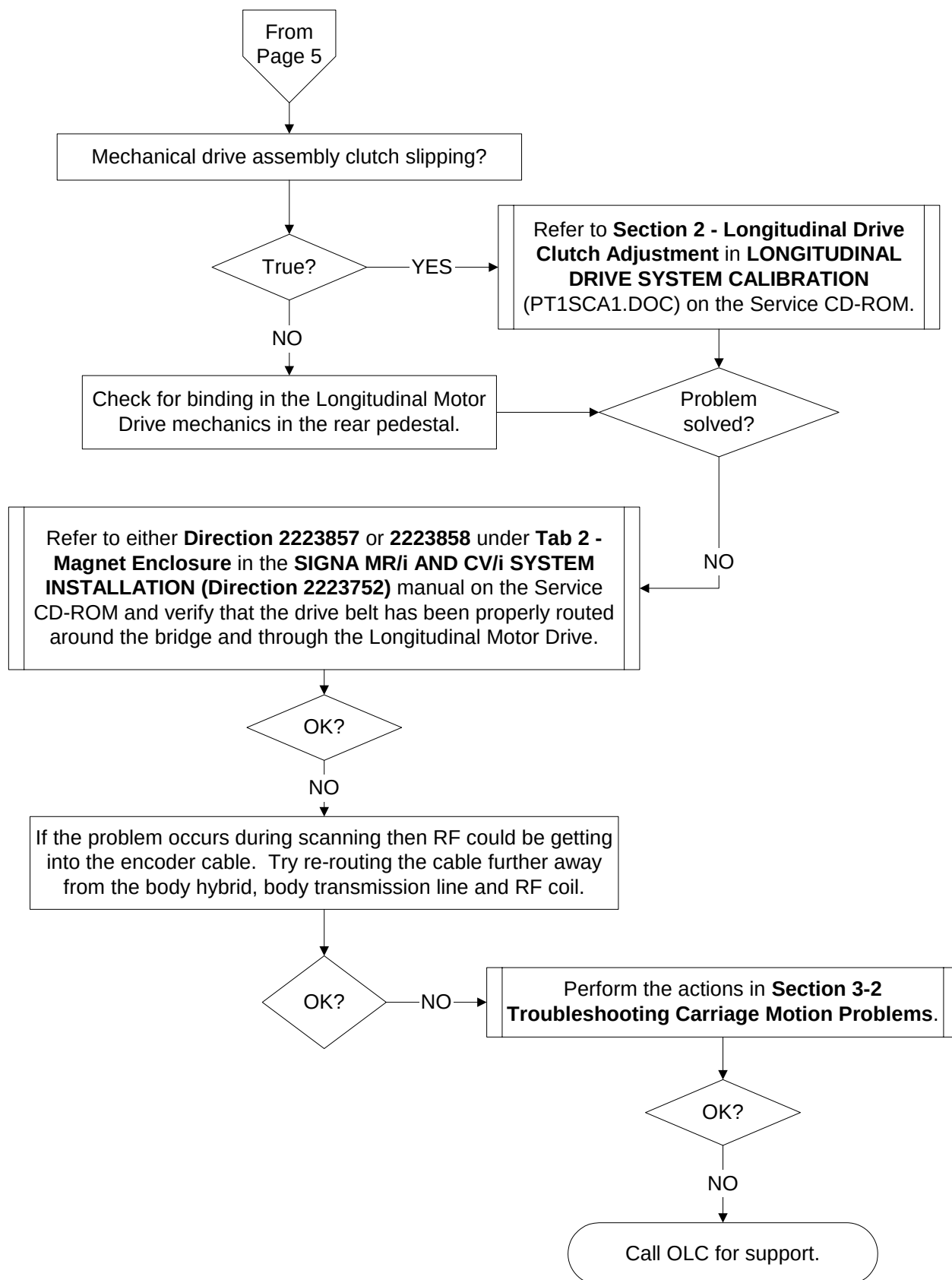
2-1 Main Body



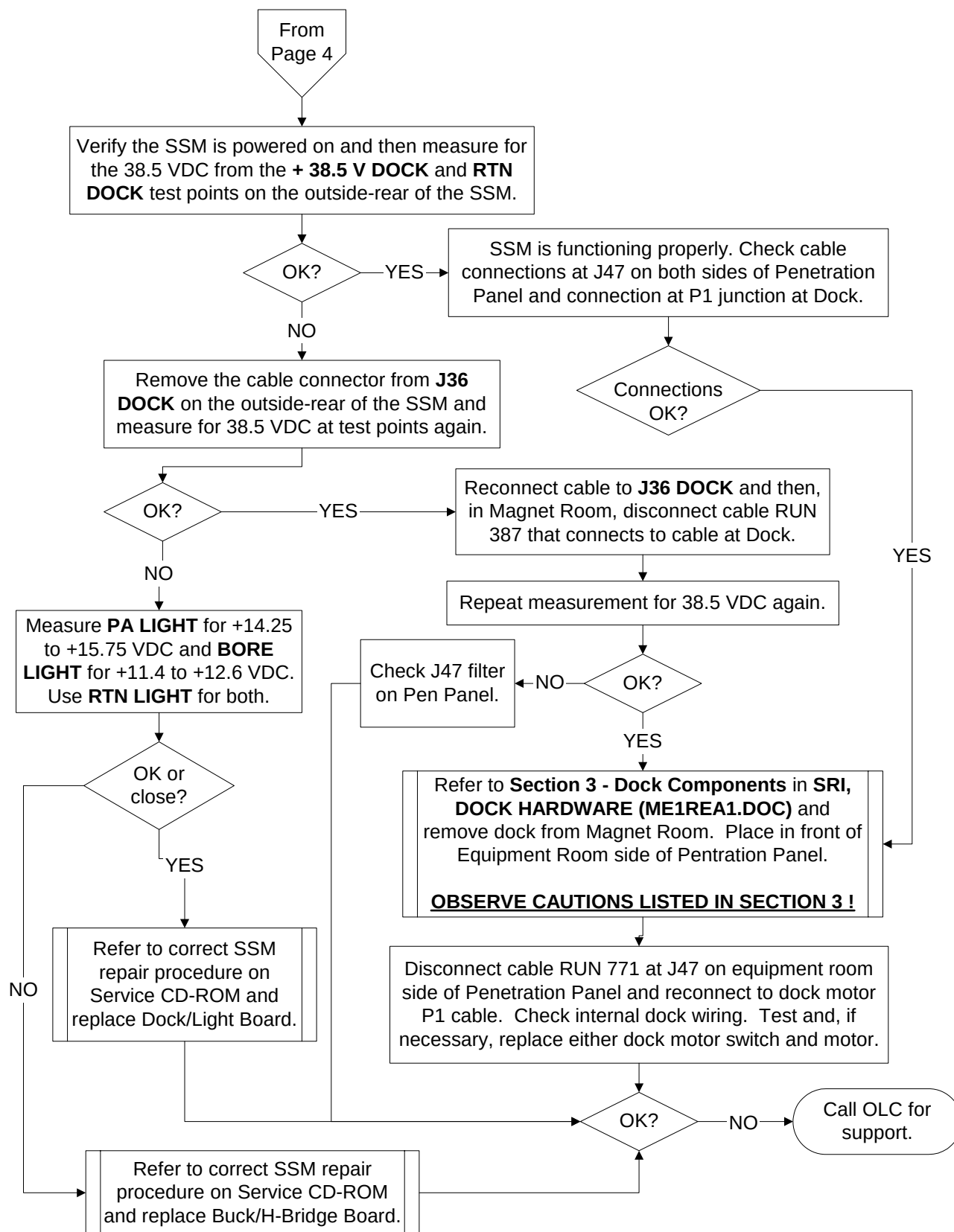
2-2 Cradle Motion Troubleshooting



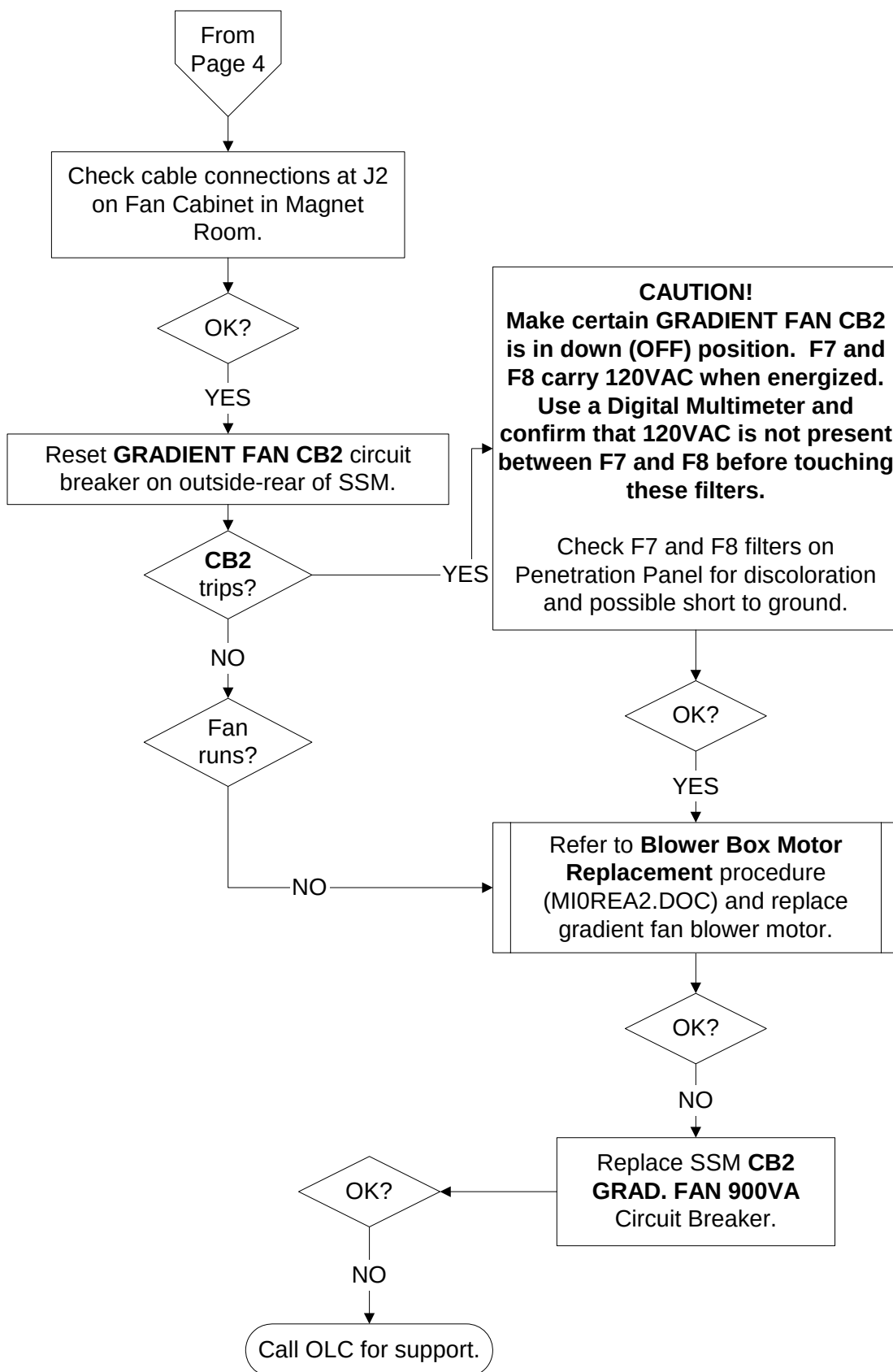
2-2 Cradle Motion Troubleshooting (Continued)



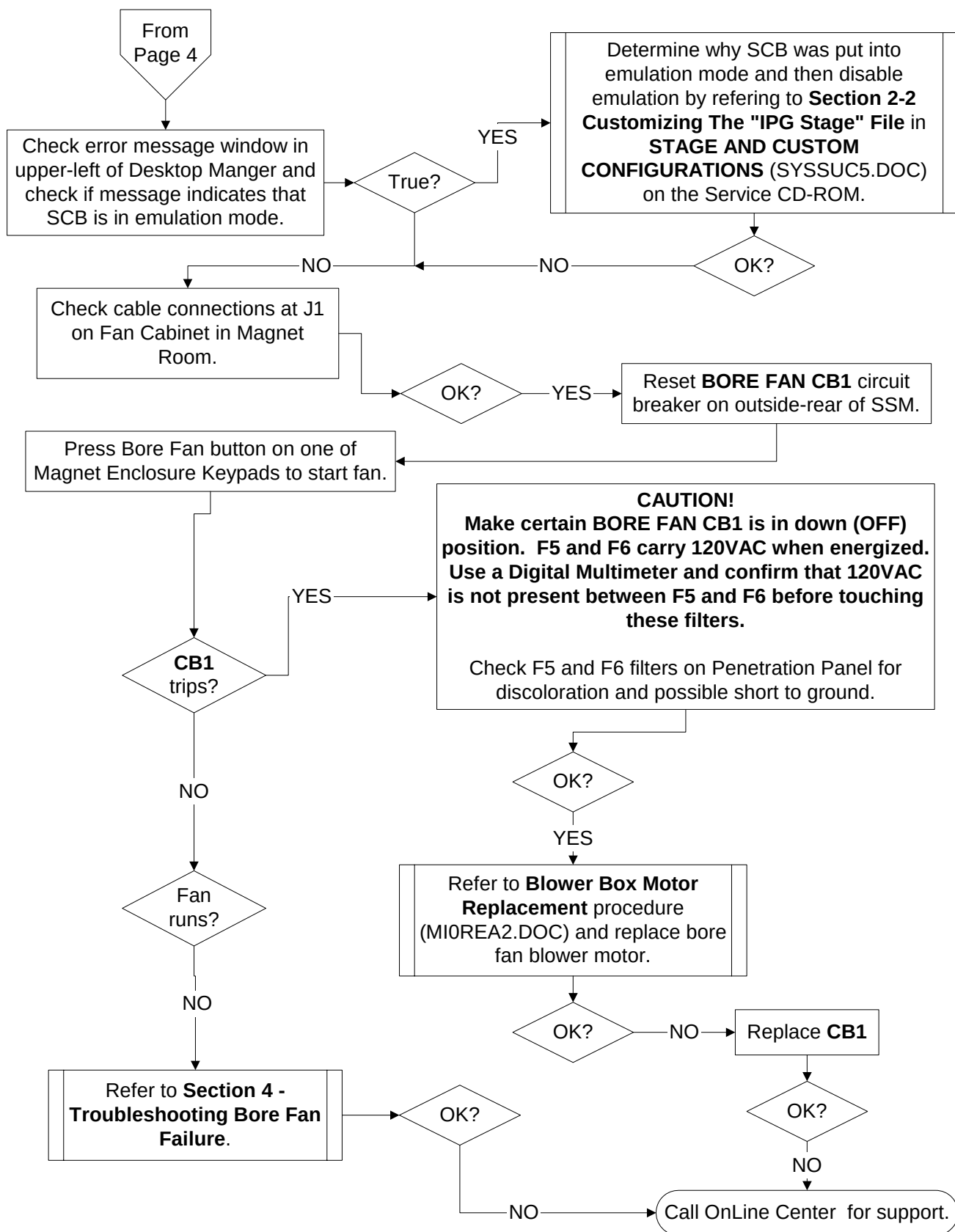
2-3 Dock Motor Troubleshooting



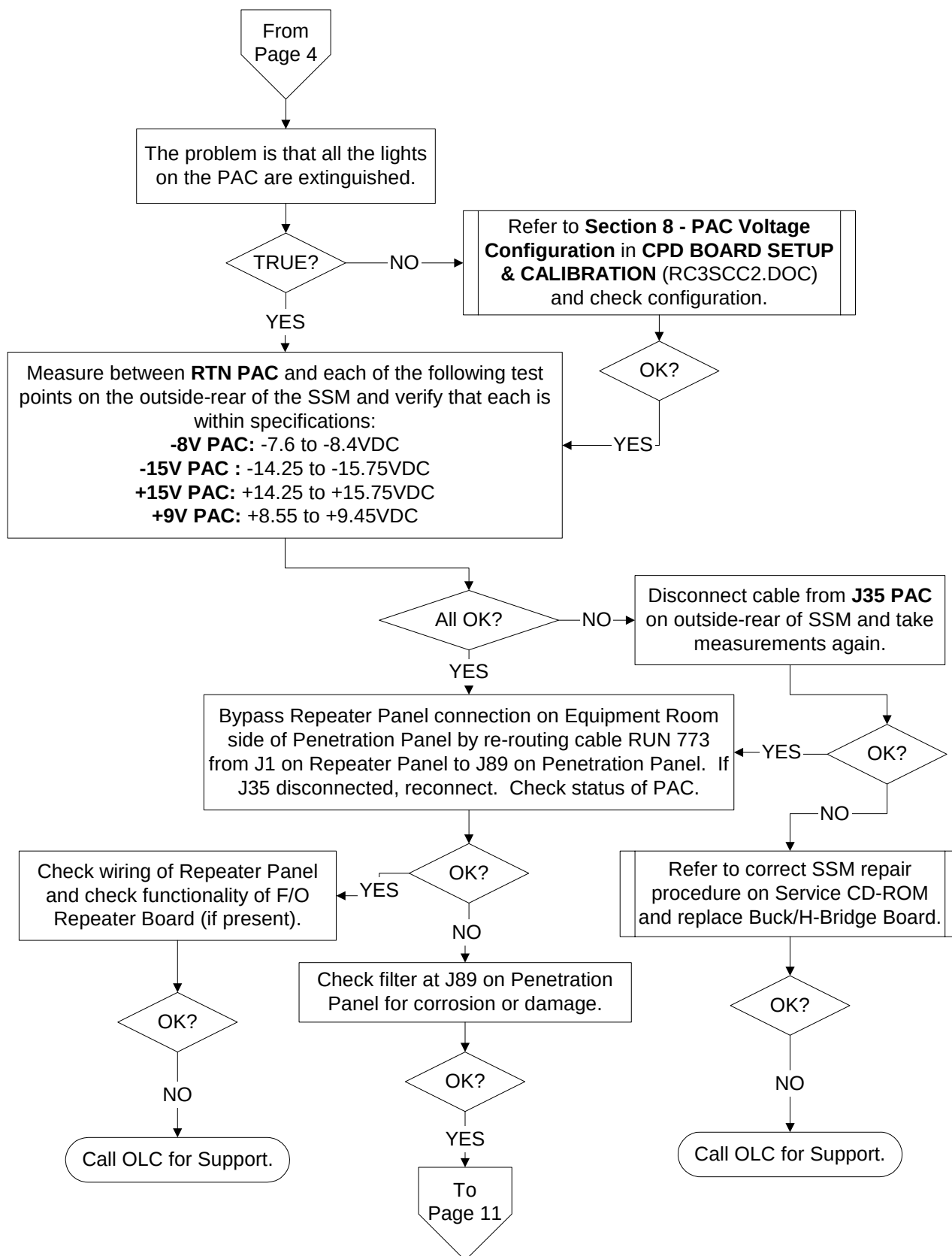
2-4 Gradinet Fan Troubleshooting



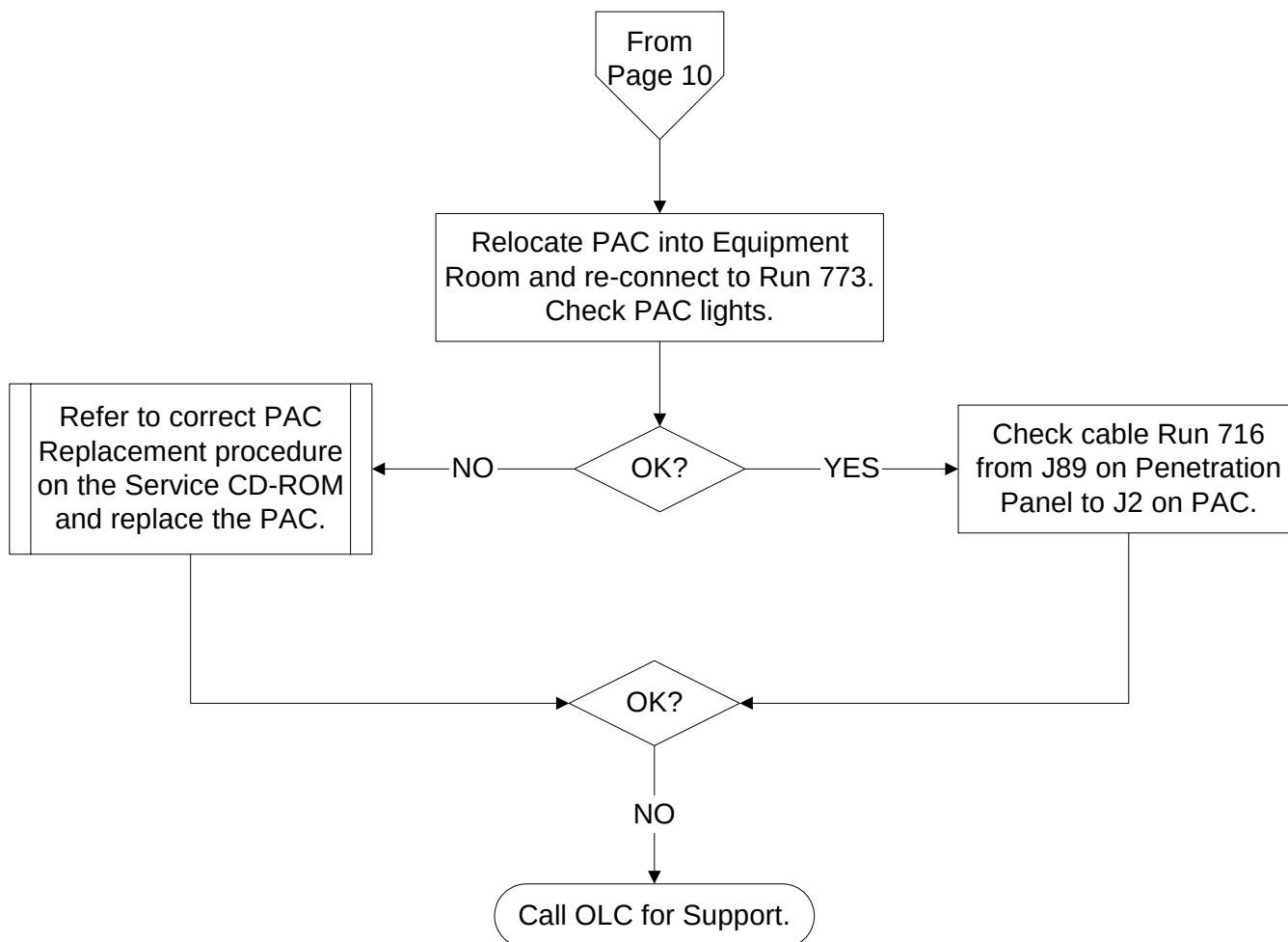
2-5 Bore Fan Troubleshooting



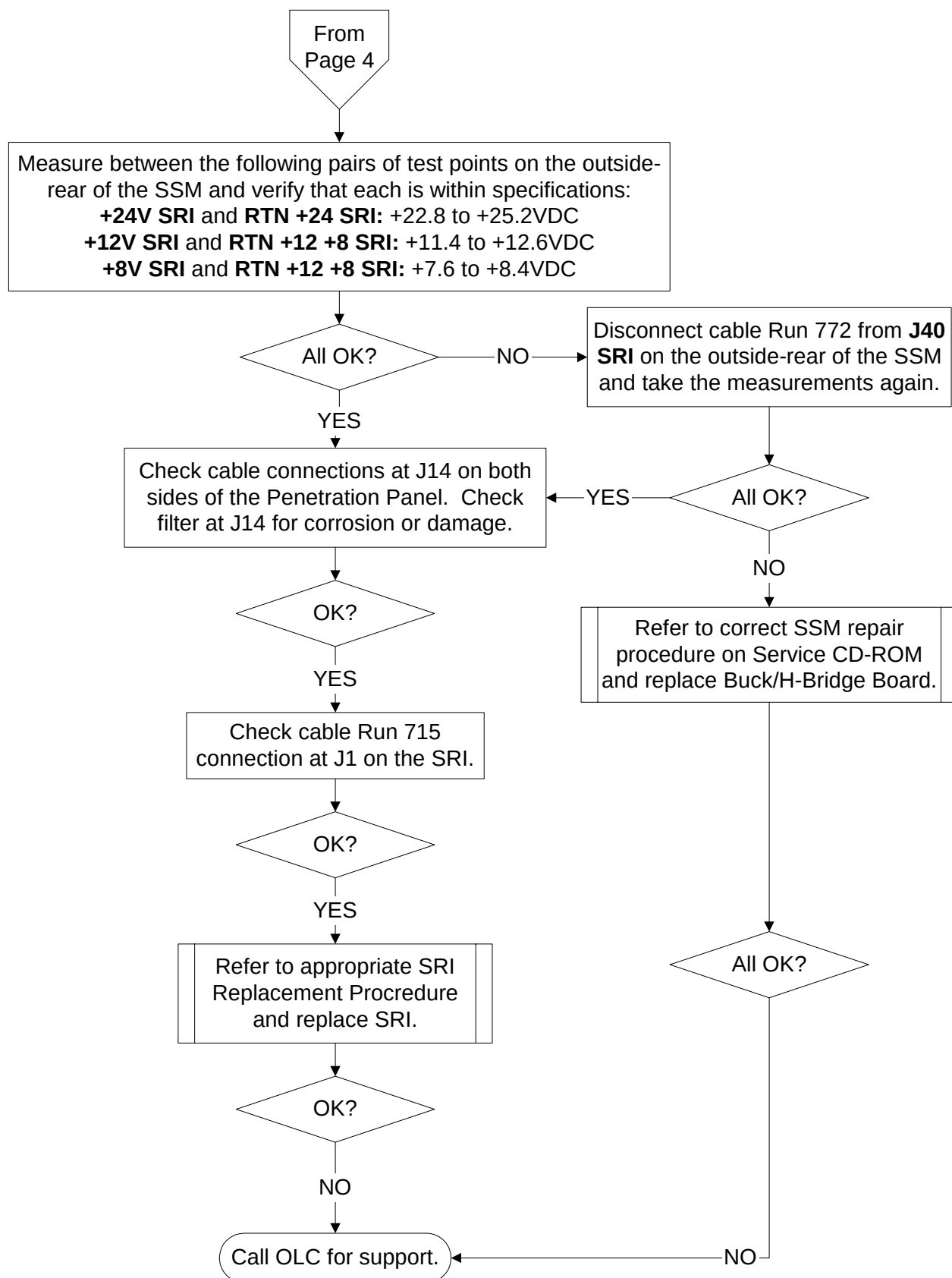
2-6 Physiological Acquisition Controller (PAC) Troubleshooting



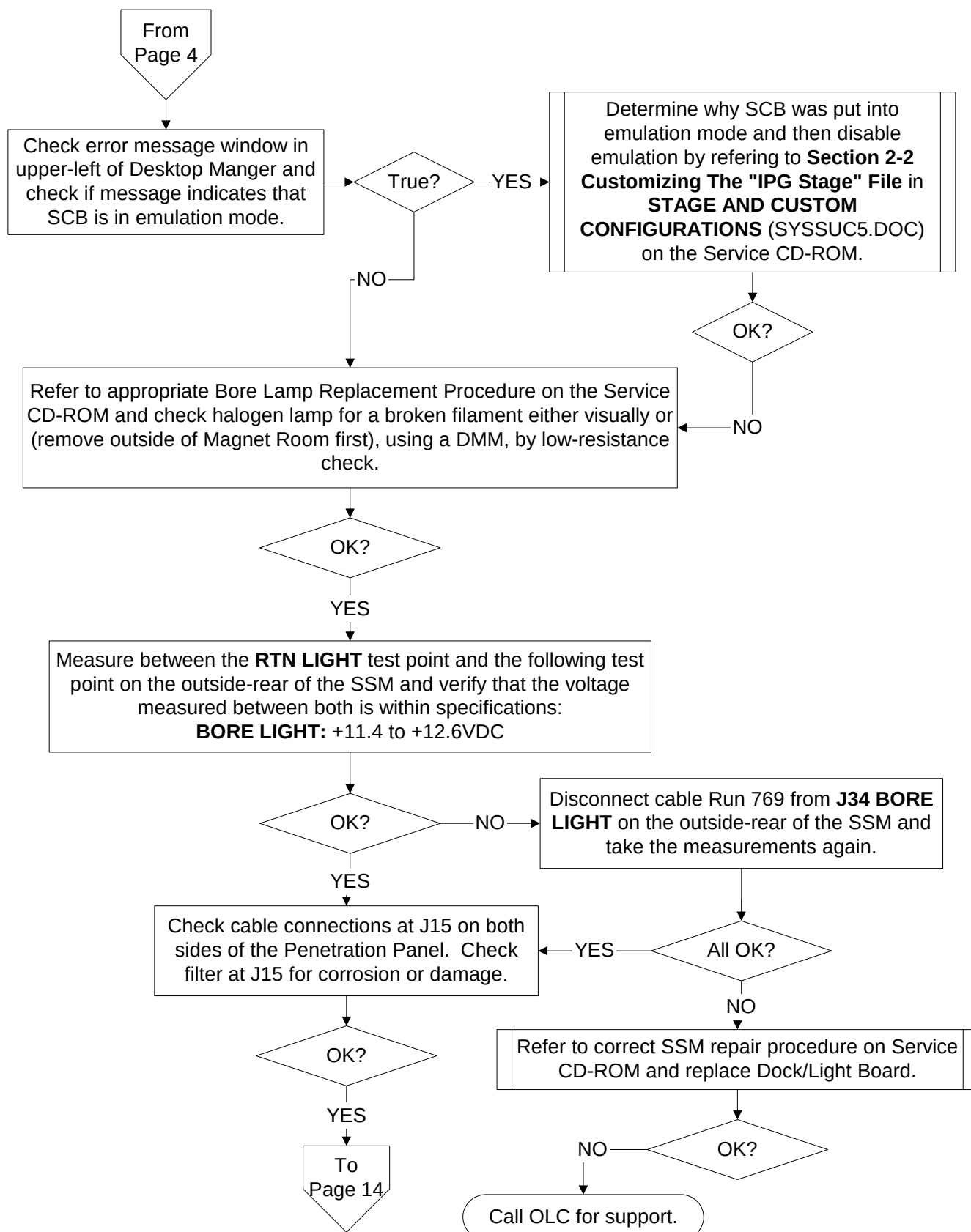
2-6 Physiological Acquisition Controller (PAC) Troubleshooting (Continued)



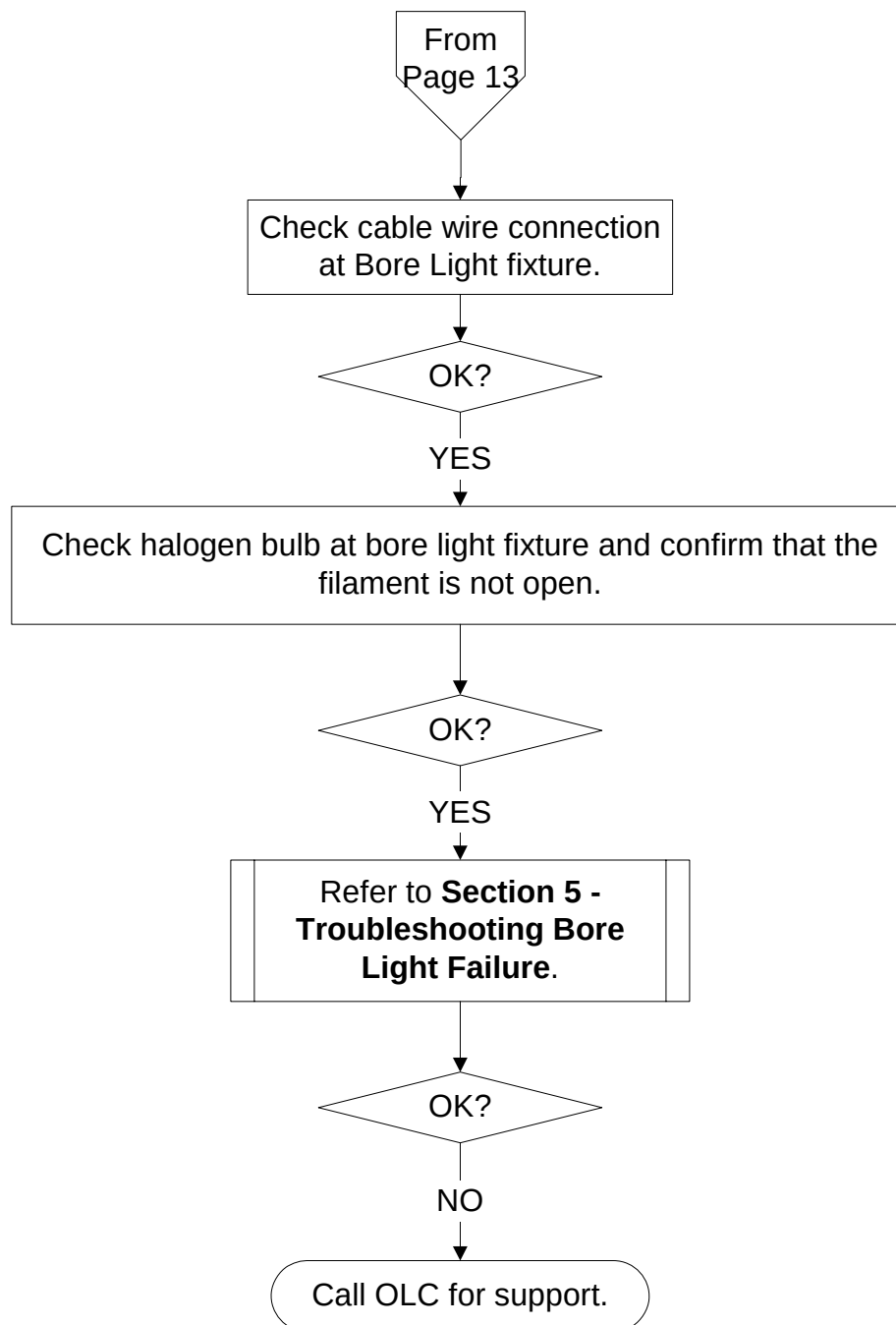
2-7 Scan Room Interface (SRI) Troubleshooting



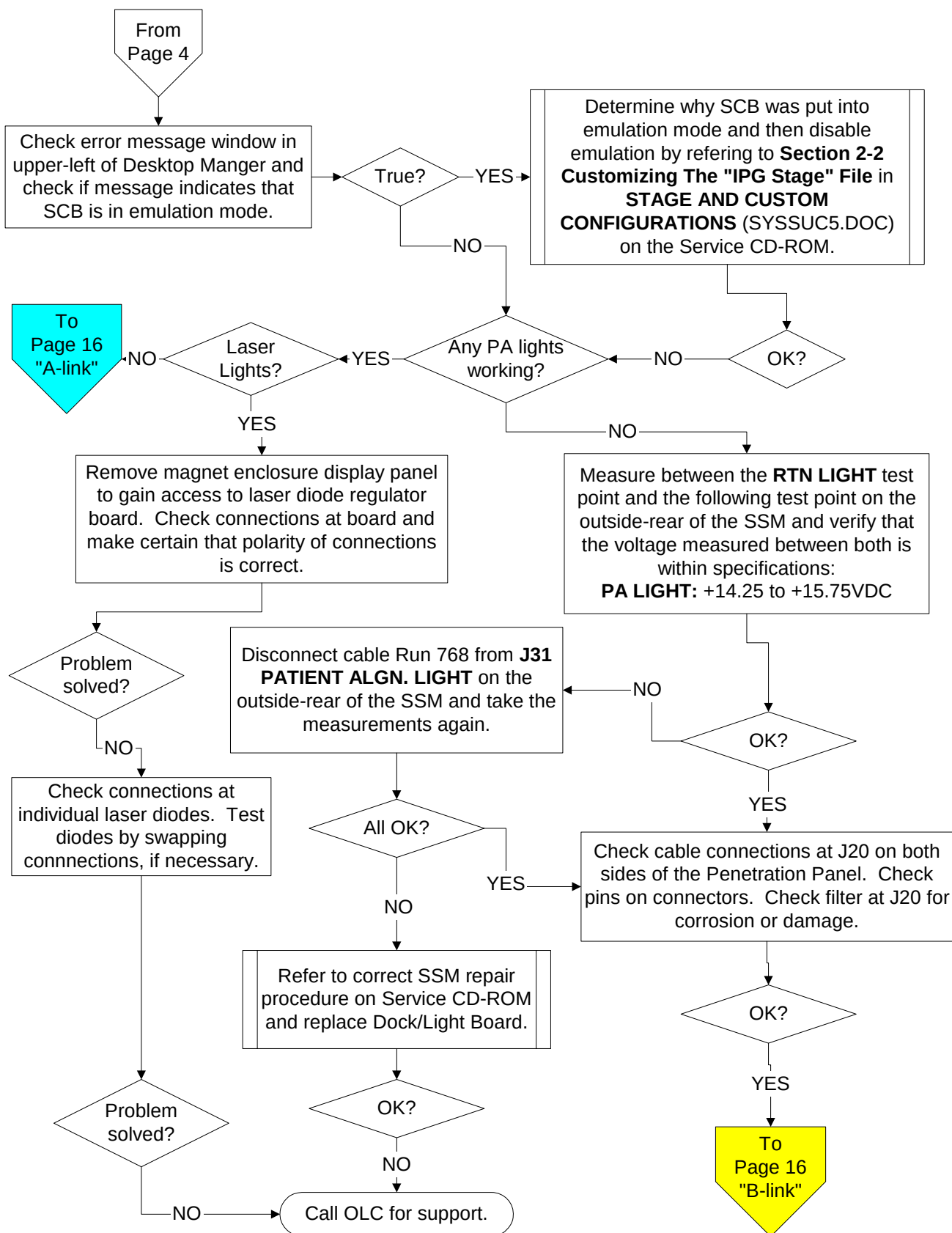
2-8 Bore Light Troubleshooting



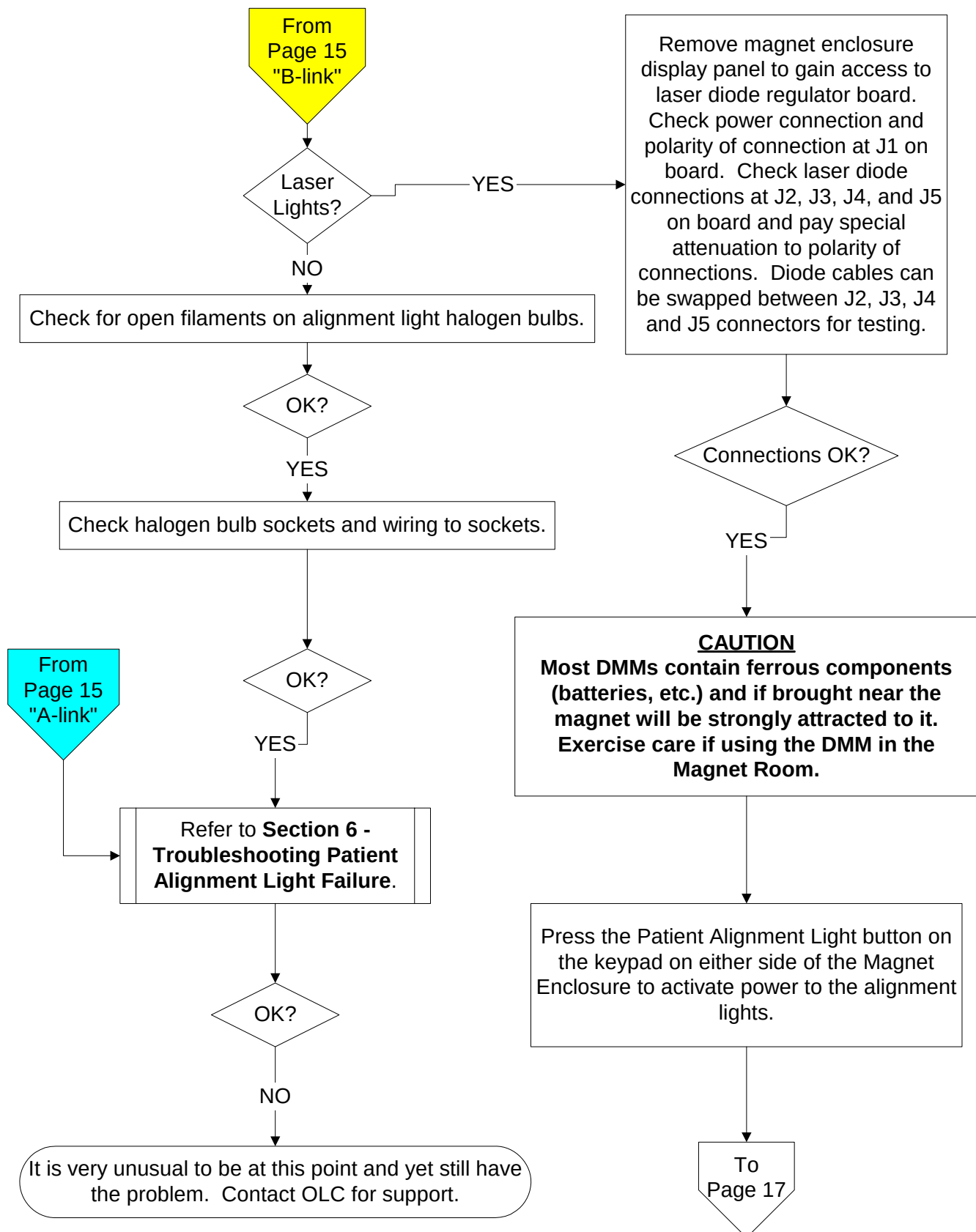
2-8 Bore Light Troubleshooting (Continued)



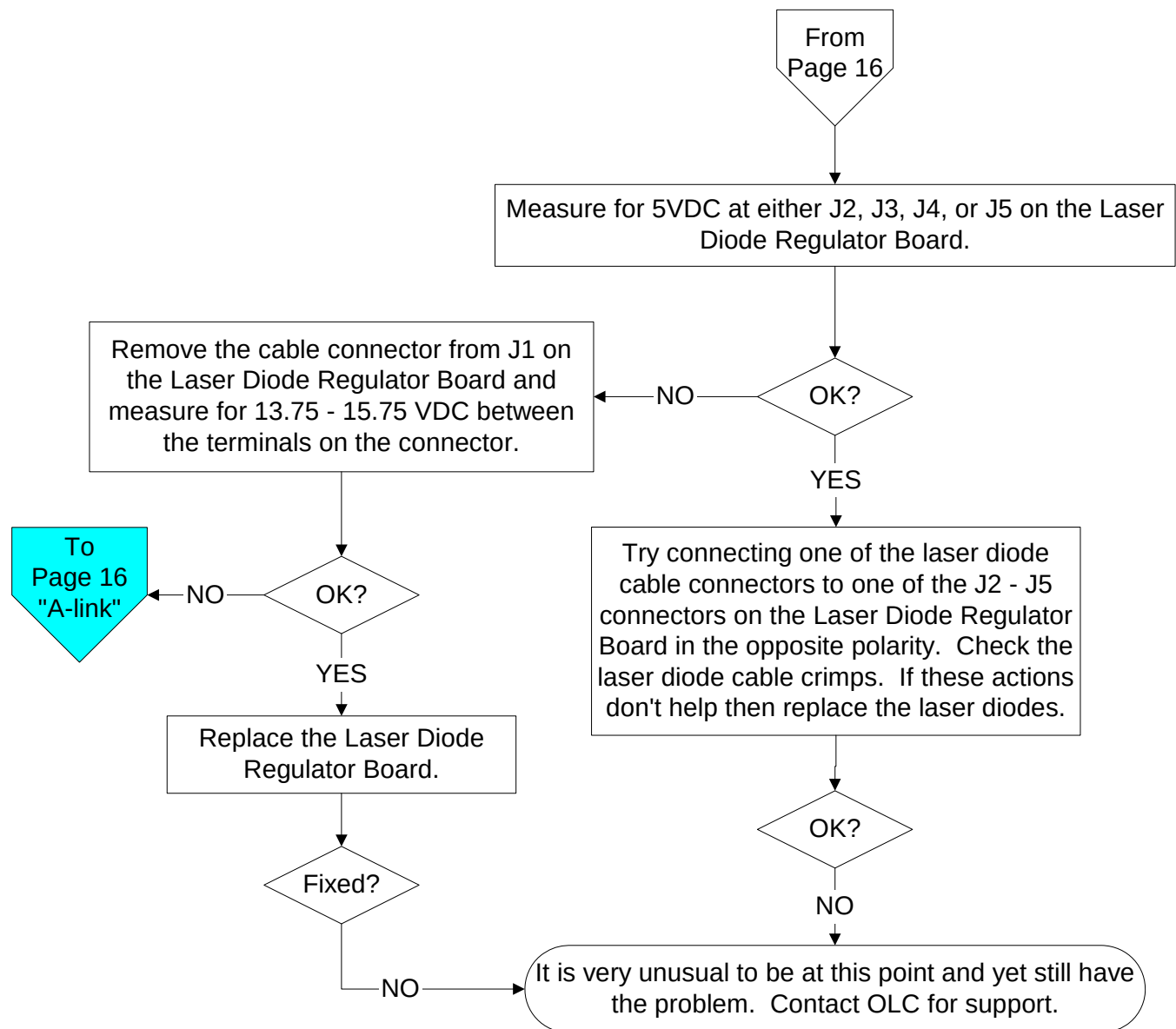
2-9 Patient Alignment Light Troubleshooting



2-9 Patient Alignment Light Troubleshooting (Continued)

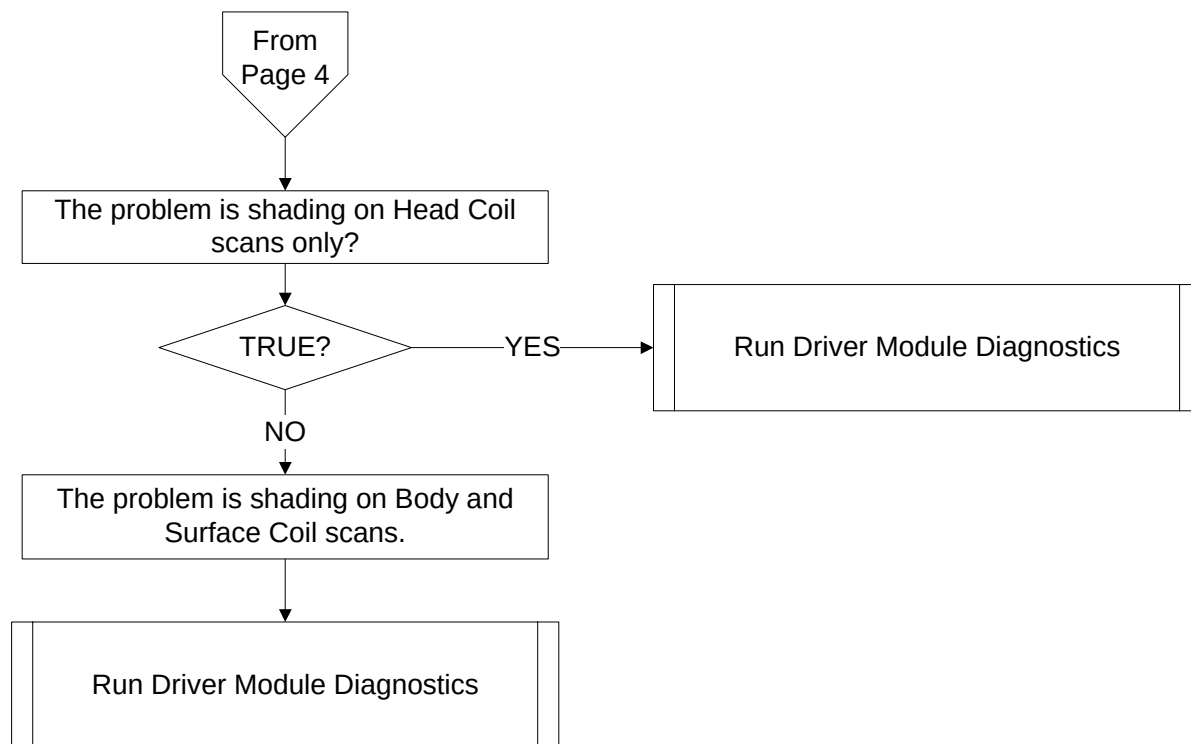


2-9 Patient Alignment Light Troubleshooting (Continued)

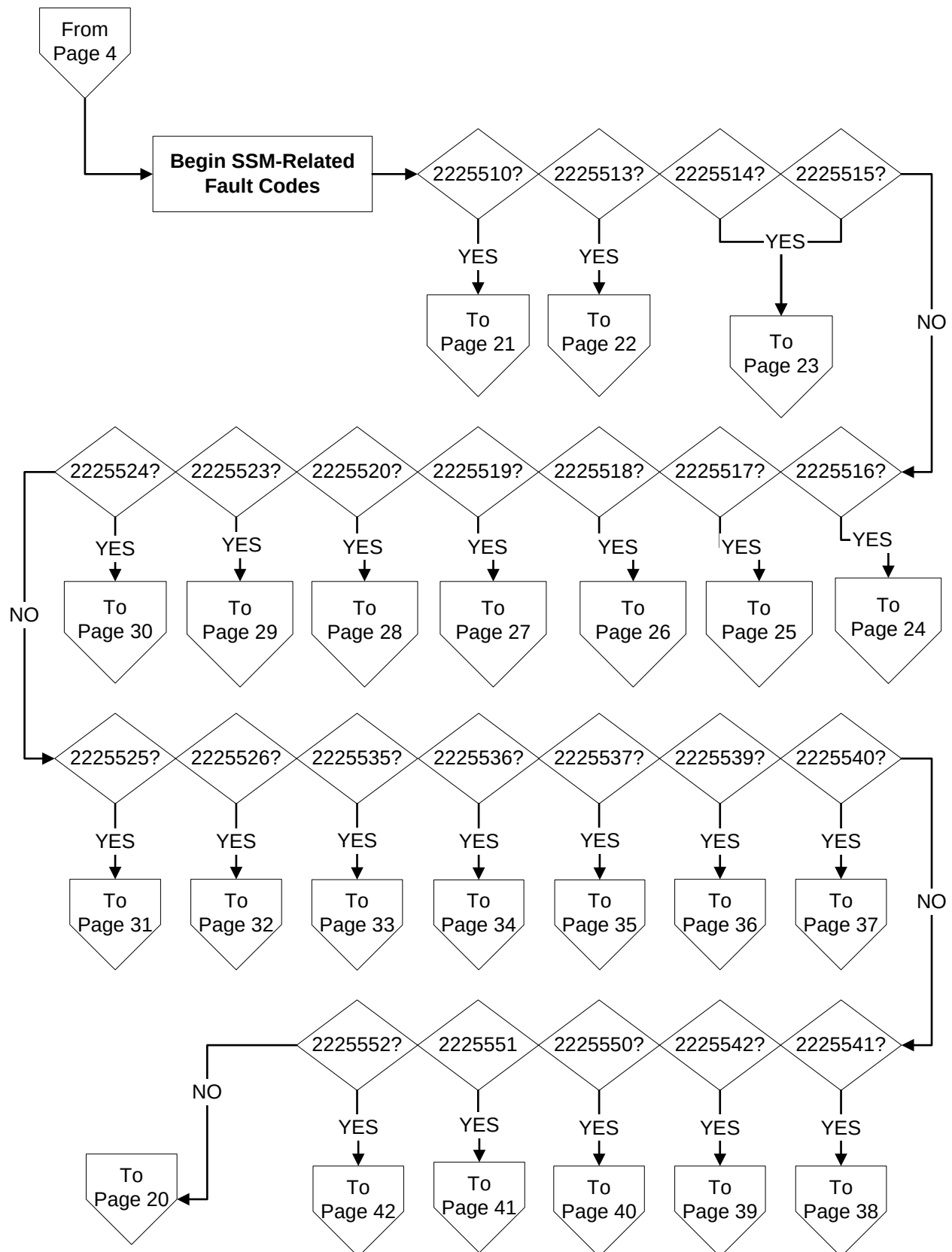


2-10 Troubleshooting Possible Image Quality Problems Related to SSM

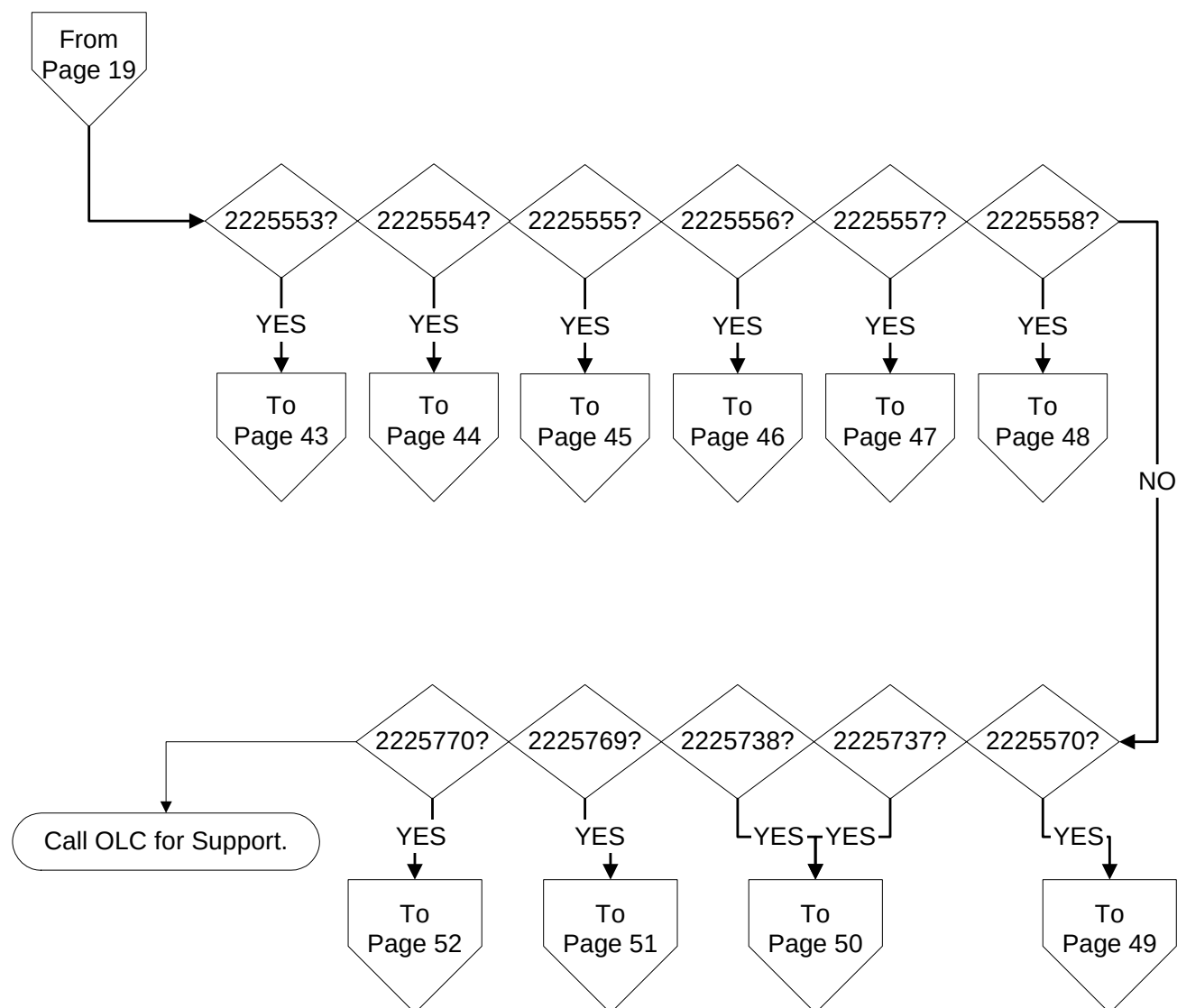
Note: The Driver Module now provides all head, body, and multicoil TR biasing functions as well as all dynamic disable and direct drive biasing.



2-11 SSM Fault Code Tree



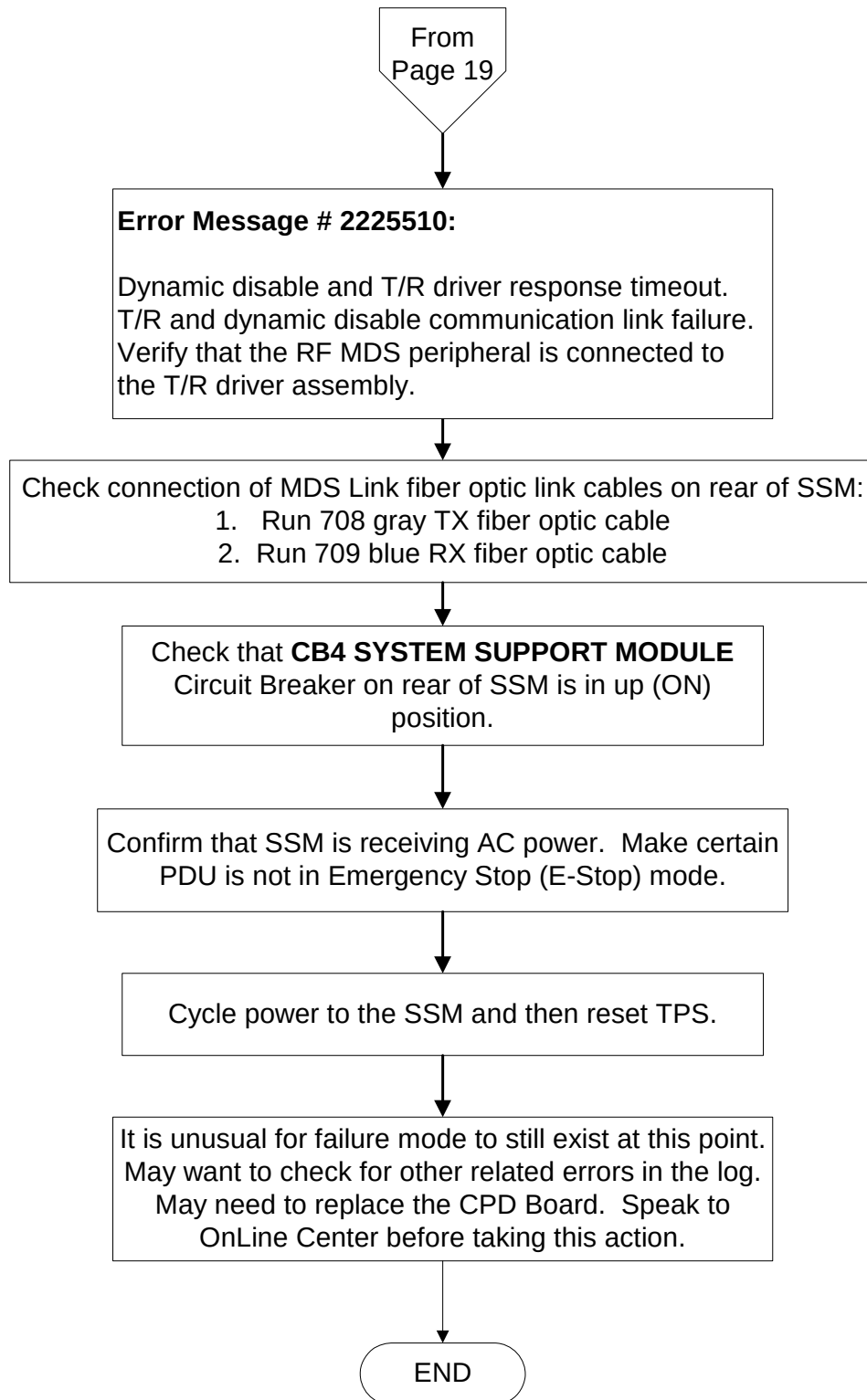
2-11 SSM Fault Code Tree (Continued)



2-12 Error Message # 2225510

Note: The Driver Module now provides all head, body, and multicoil TR biasing functions as well as all dynamic disable and direct drive biasing.

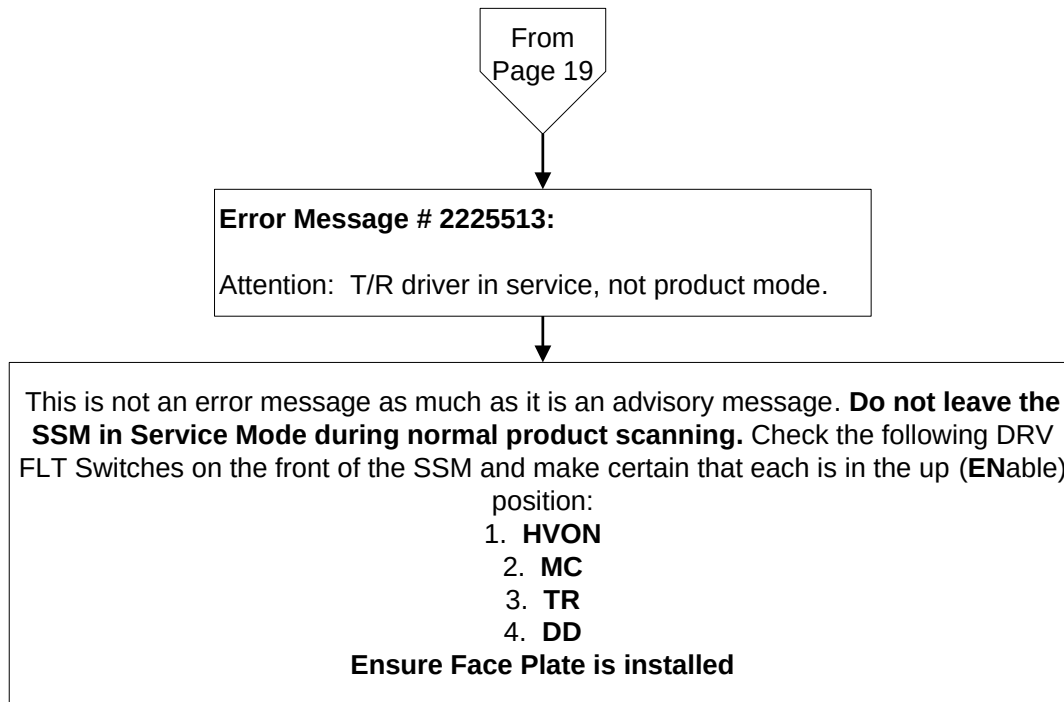
Note: Under normal conditions you should not see these error codes. The SSM no longer performs these functions.



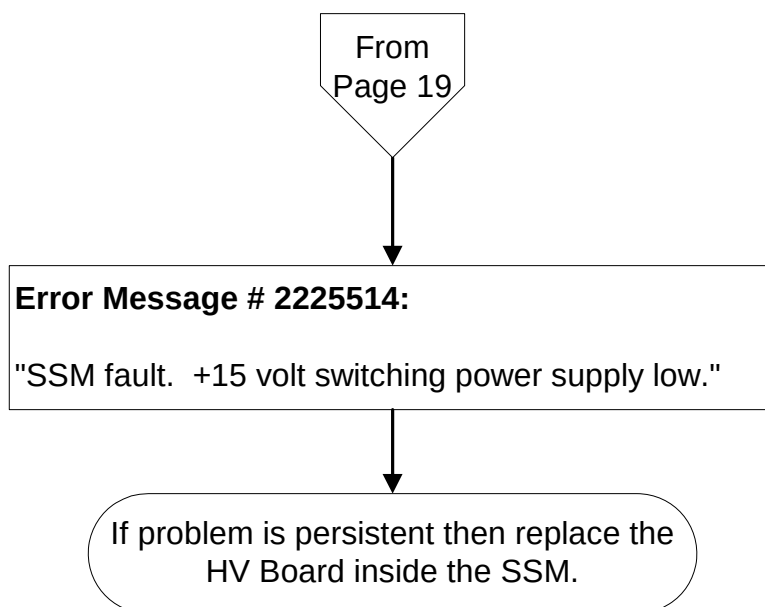
2-13 Error Message # 2225513

Note: The Driver Module now provides all head, body, and multicoil TR biasing functions as well as all dynamic disable and direct drive biasing.

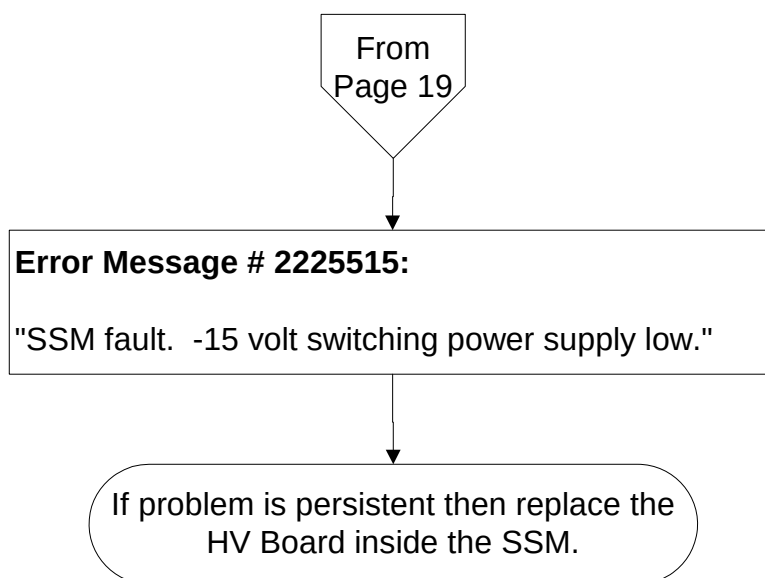
Note: Under normal conditions you should not see these error codes. The SSM no longer performs these functions.



2-14 Error Message # 2225514



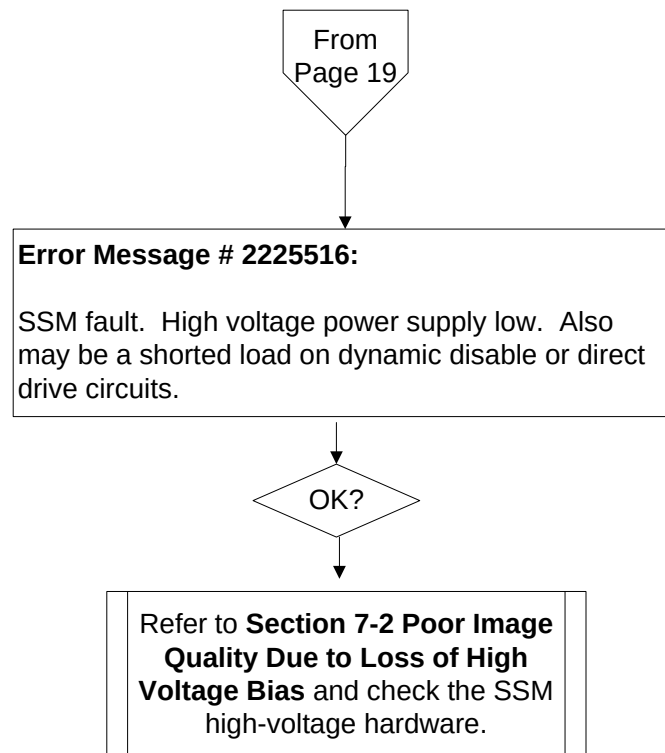
2-15 Error Message # 2225515



Note: HV on the SSM is not used in the system. HV is provided by the Driver Module.

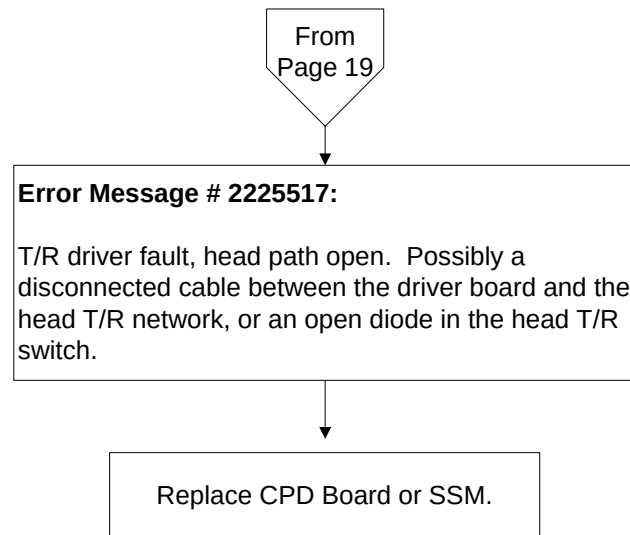
2-16 Error Message # 2225516

Note: HV on the SSM is not used in the system. HV is provided by the Driver Module.



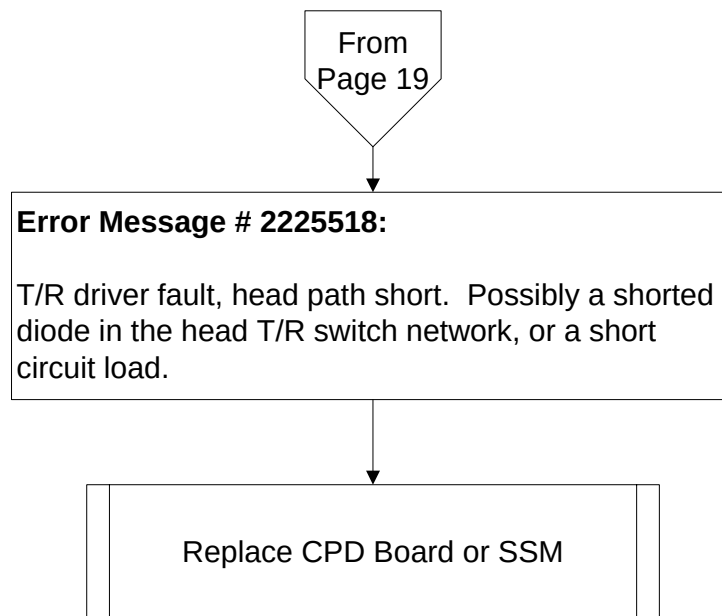
2-17 Error Message # 2225517

Note: Under normal conditions you should not see this error code. The SSM no longer performs these functions.



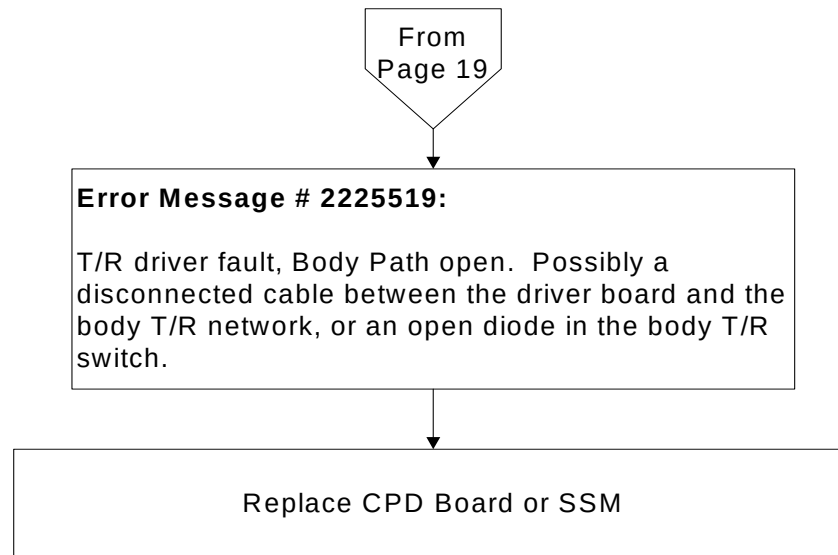
2-18 Error Message # 2225518

Note: Under normal conditions you should not see this error code. The SSM no longer performs these functions.



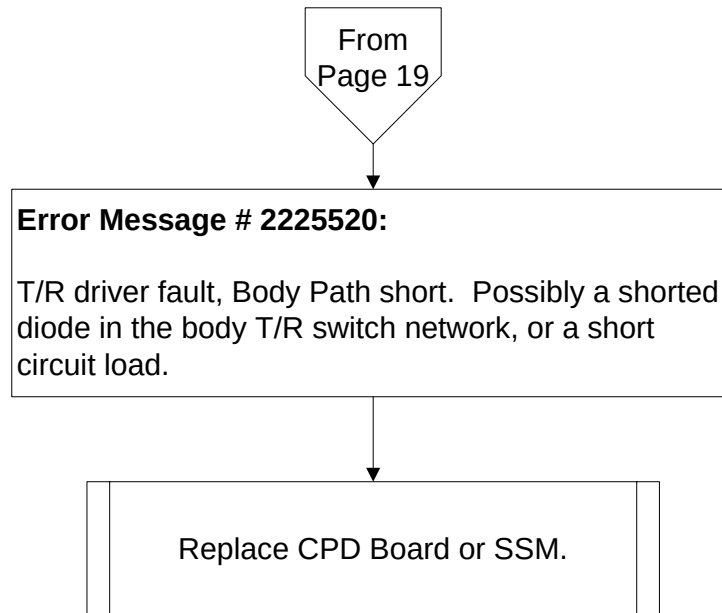
2-19 Error Message # 2225519

Note: Under normal conditions you should not see this error code. The SSM no longer performs these functions.



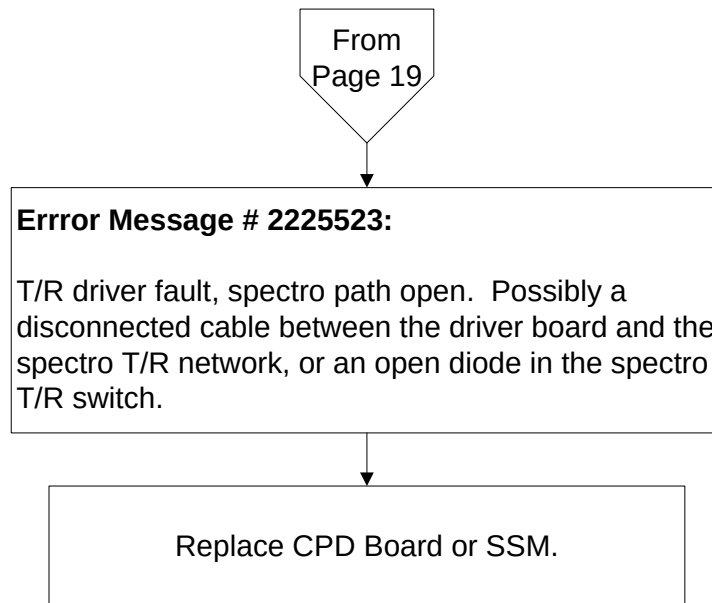
2-20 Error Message # 2225520

Note: Under normal conditions you should not see this error code. The SSM no longer performs these functions.



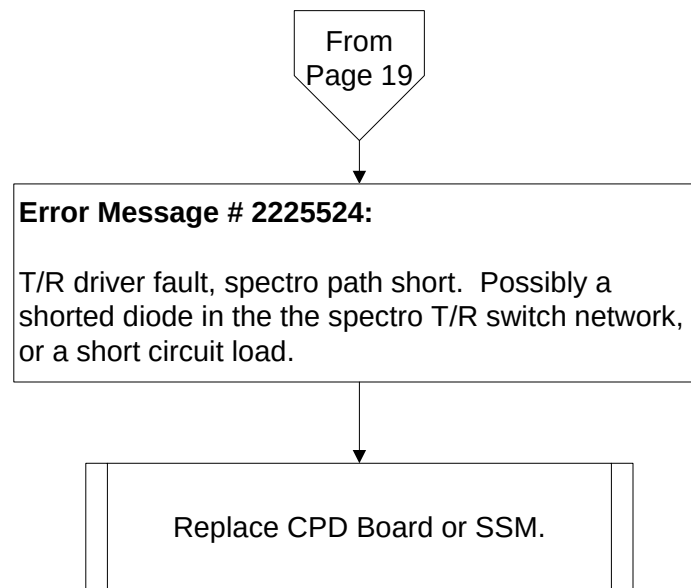
2-21 Error Message # 2225523

Note: Under normal conditions you should not see this error code. The SSM no longer performs these functions.



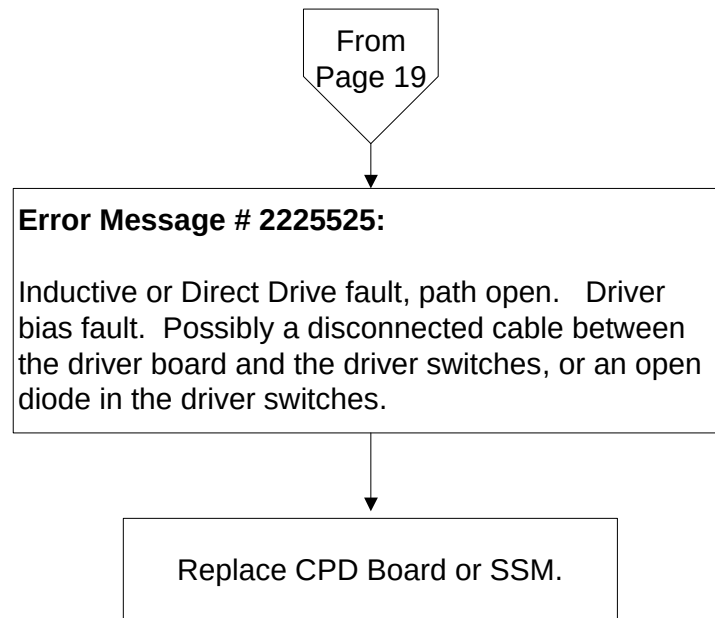
2-22 Error Message # 2225524

Note: Under normal conditions you should not see this error code. The SSM no longer performs these functions.



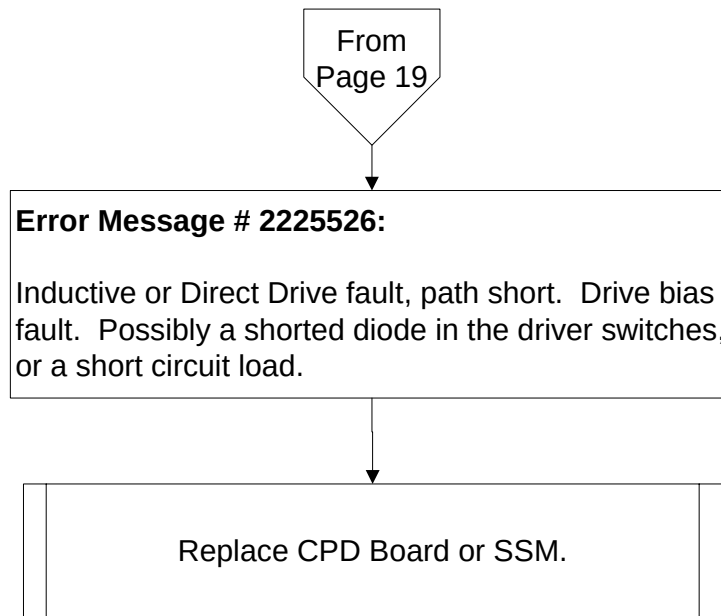
2-23 Error Message # 2225525

Note: Under normal conditions you should not see this error code. The SSM no longer performs these functions.



2-24 Error Message # 2225526

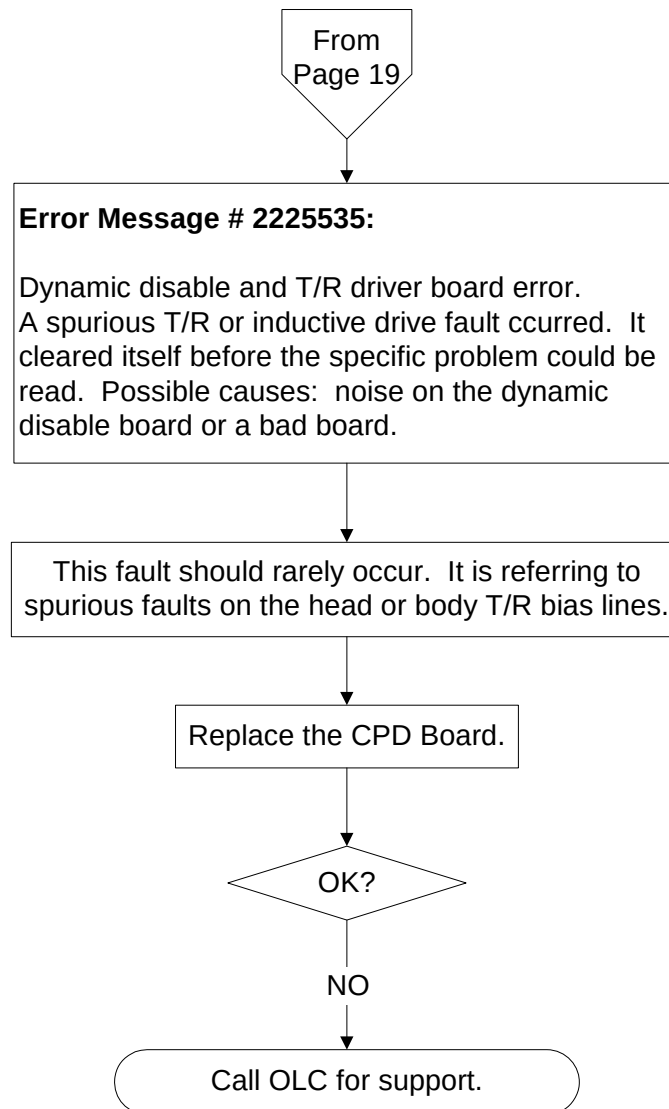
Note: Under normal conditions you should not see this error code. The SSM no longer performs these functions.



2-25 Error Message # 2225535

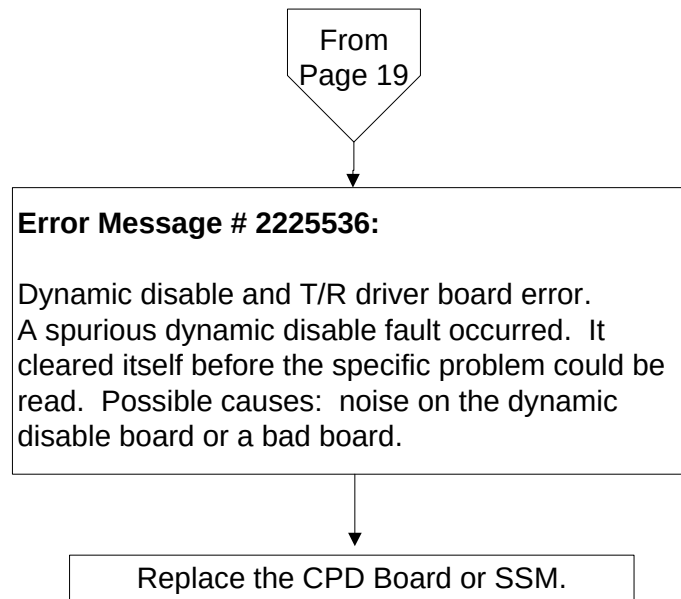
Note: The Driver Module now provides all head, body, and multicoil TR biasing functions as well as all dynamic disable and direct drive biasing.

Note: Under normal conditions you should not see this error code. The SSM no longer performs these functions.



2-26 Error Message # 2225536

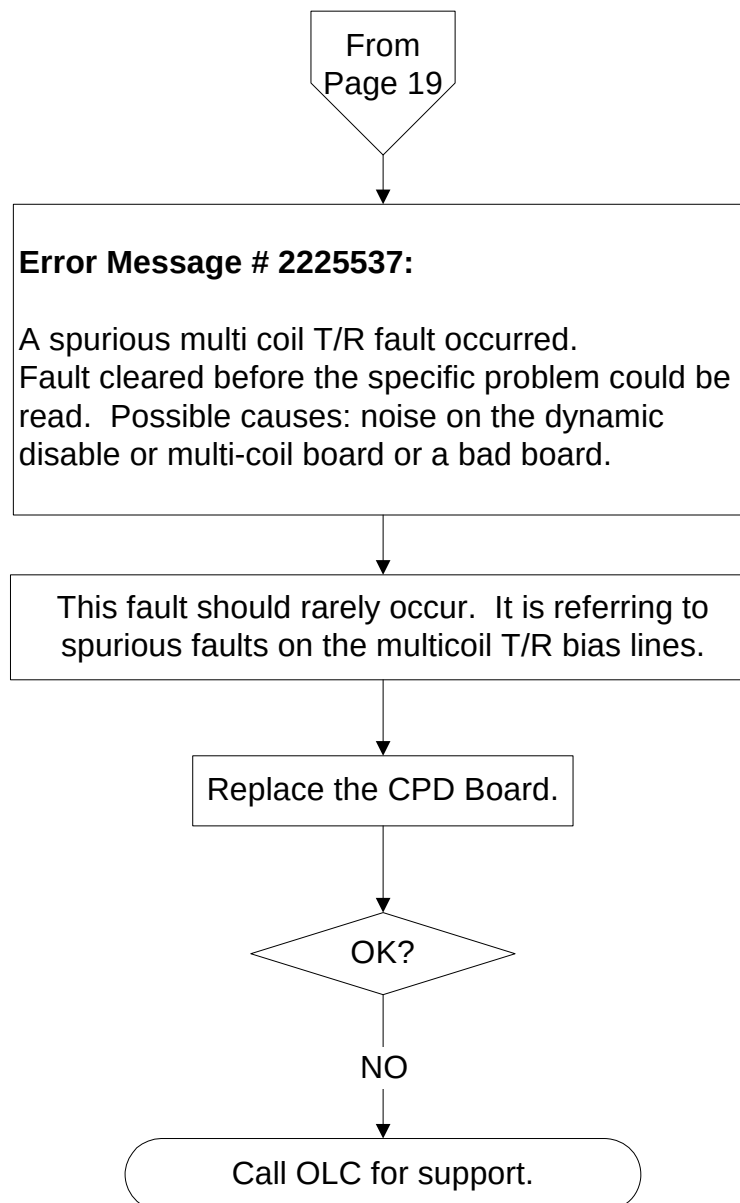
Note: Under normal conditions you should not see this error code. The SSM no longer performs these functions.



2-27 Error Message # 2225537

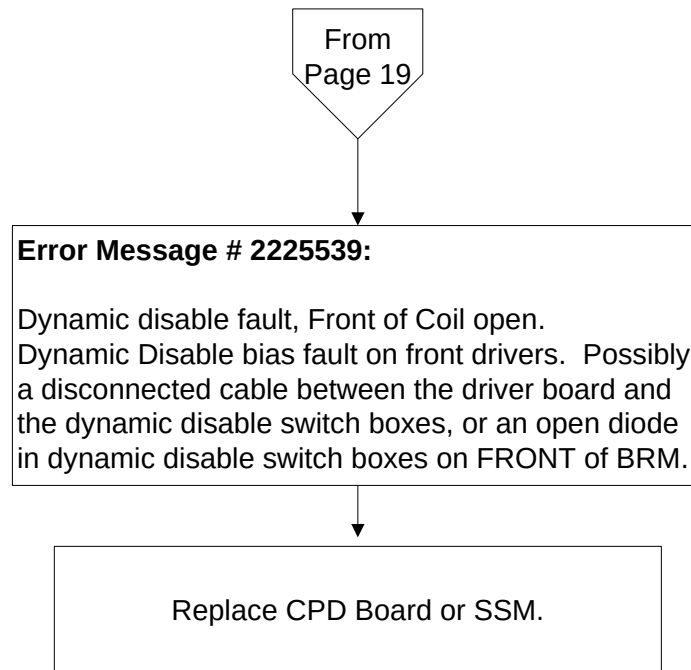
Note: The Driver Module now provides all head, body, and multicoil TR biasing functions as well as all dynamic disable and direct drive biasing.

Note: Under normal conditions you should not see this error code. The SSM no longer performs these functions.



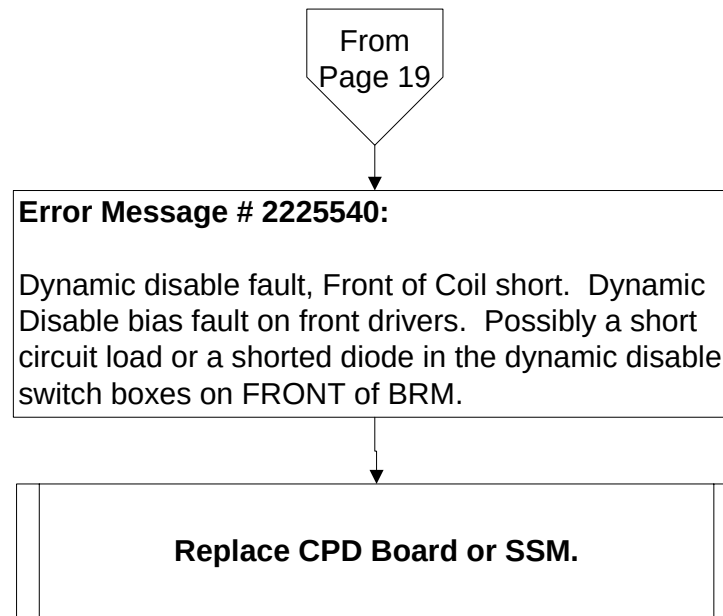
2-28 Error Message # 2225539

Note: Under normal conditions you should not see this error code. The SSM no longer performs these functions.



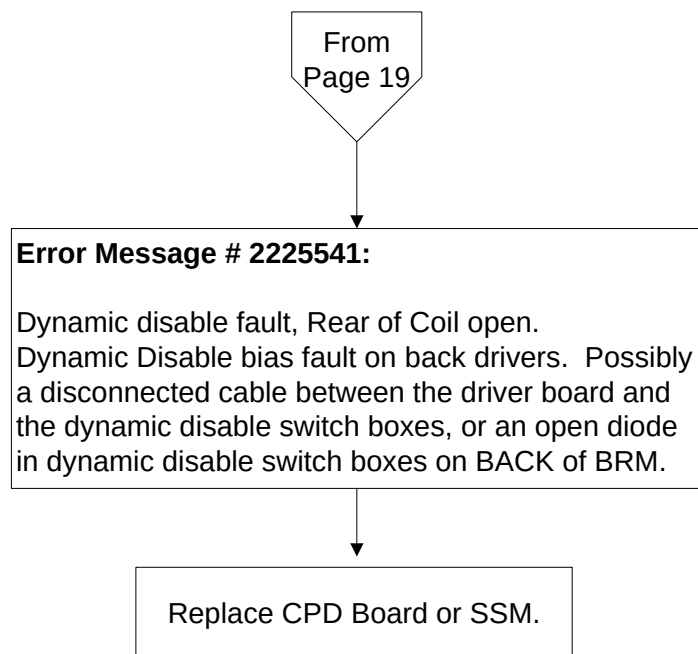
2-29 Error Message # 2225540

Note: Under normal conditions you should not see this error code. The SSM no longer performs these functions.



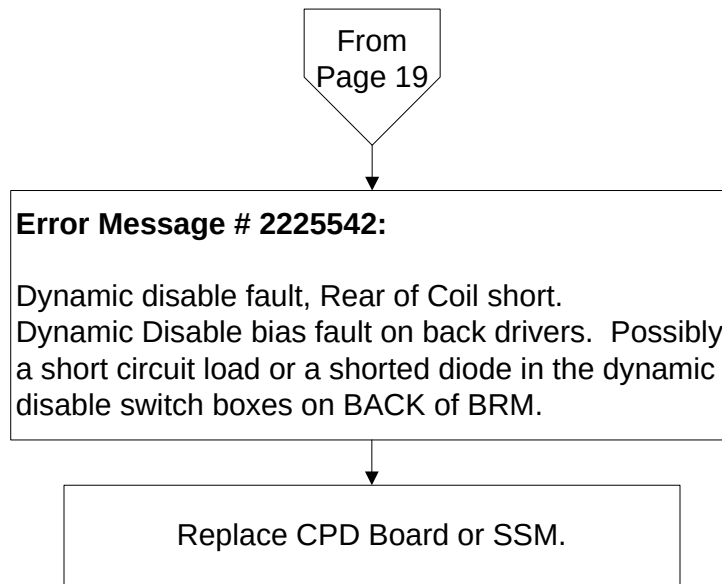
2-30 Error Message # 2225541

Note: Under normal conditions you should not see this error code. The SSM no longer performs these functions.



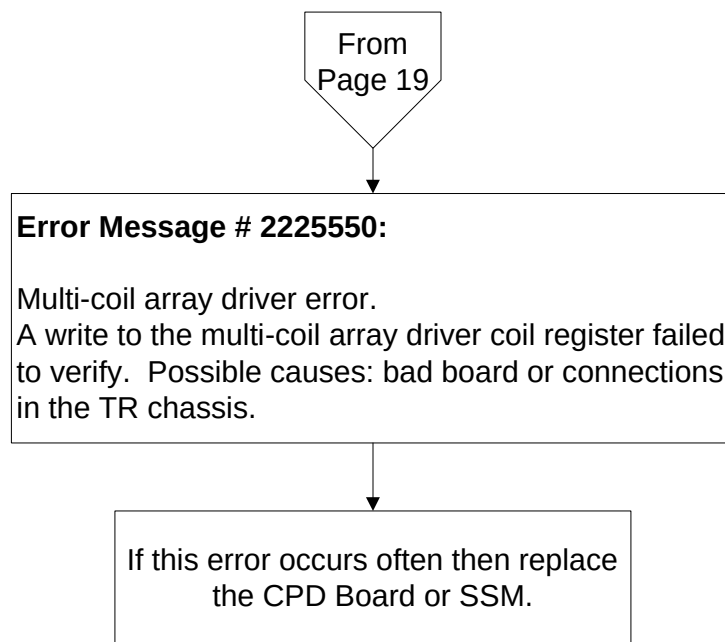
2-31 Error Message # 2225542

Note: Under normal conditions you should not see this error code. The SSM no longer performs these functions.



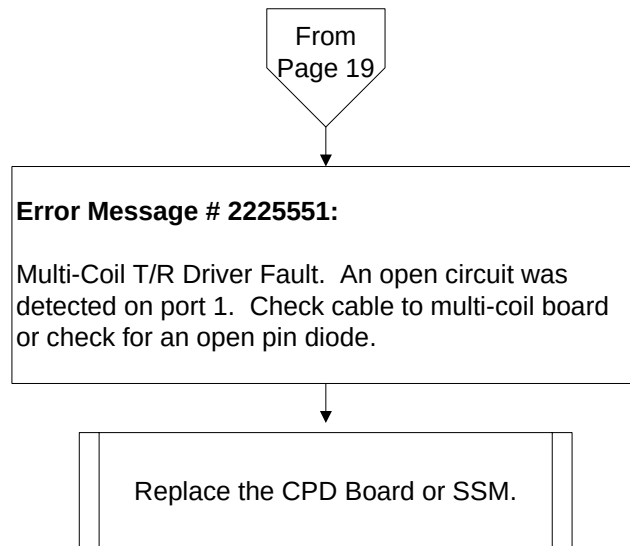
2-32 Error Message # 2225550

Note: Under normal conditions you should not see this error code. The SSM no longer performs these functions.



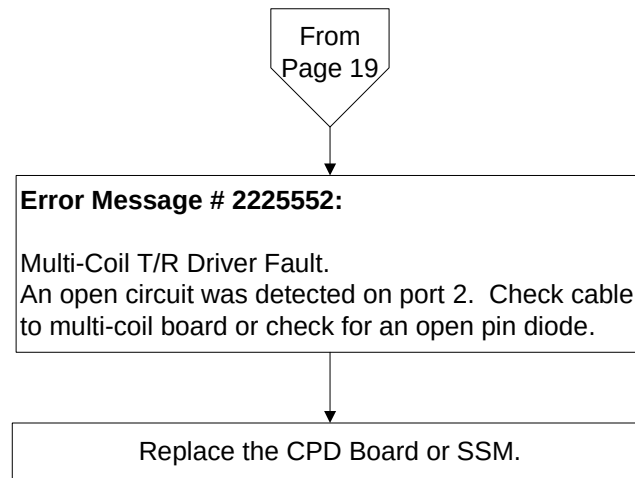
2-33 Error Message # 2225551

Note: Under normal conditions you should not see this error code. The SSM no longer performs these functions.



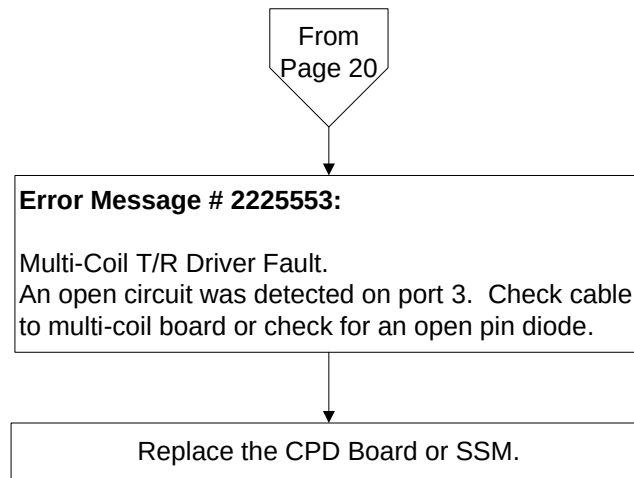
2-34 Error Message # 2225552

Note: Under normal conditions you should not see this error code. The SSM no longer performs these functions.



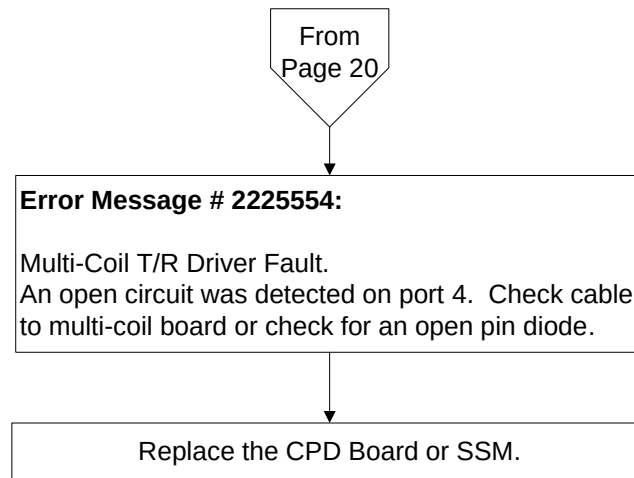
2-35 Error Message # 2225553

Note: Under normal conditions you should not see this error code. The SSM no longer performs these functions.



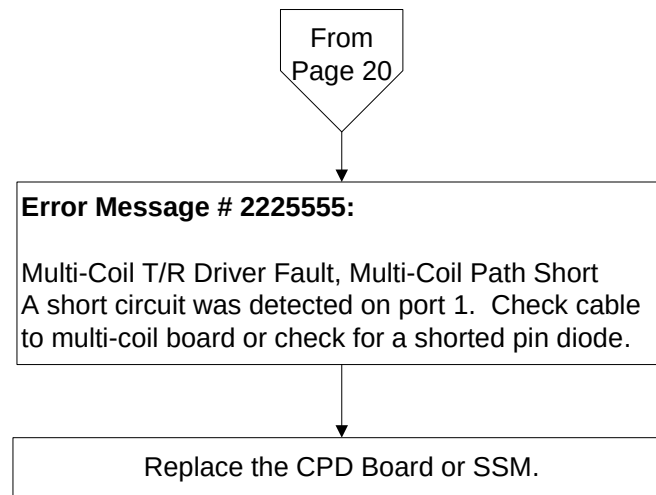
2-36 Error Message # 2225554

Note: Under normal conditions you should not see this error code. The SSM no longer performs these functions.



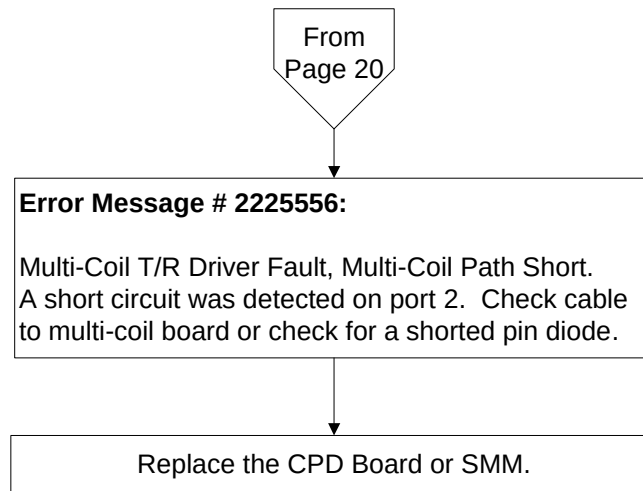
2-37 Error Message # 2225555

Note: Under normal conditions you should not see this error code. The SSM no longer performs these functions.



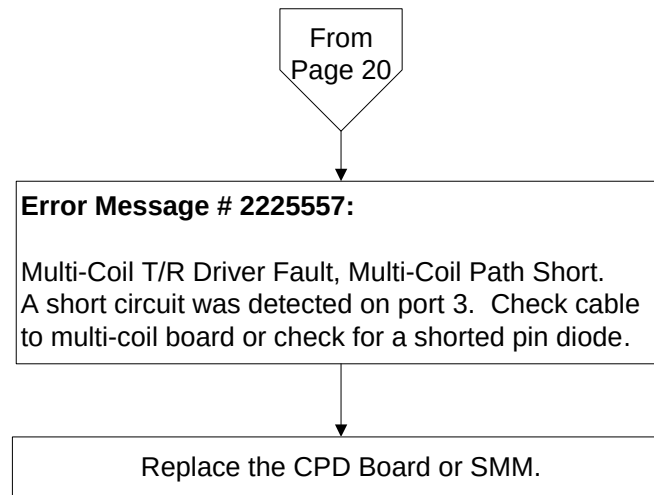
2-38 Error Message # 2225556

Note: Under normal conditions you should not see this error code. The SSM no longer performs these functions.



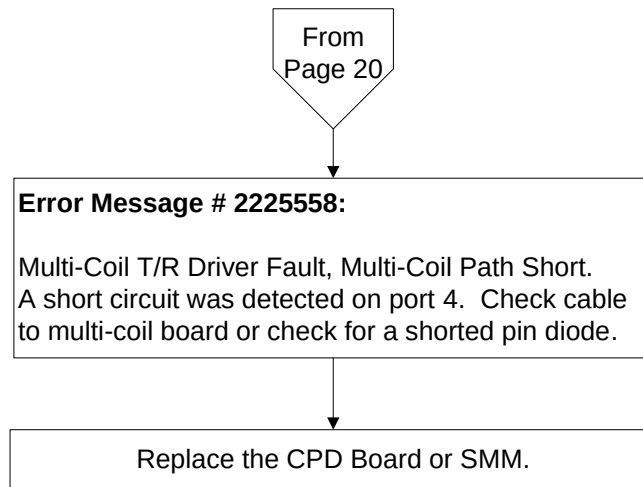
2-39 Error Message # 2225557

Note: Under normal conditions you should not see this error code. The SSM no longer performs these functions.

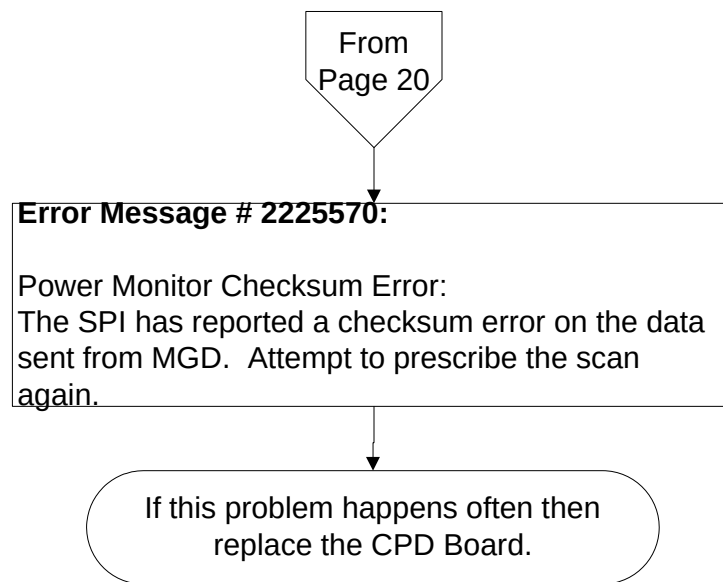


2-40 Error Message # 2225558

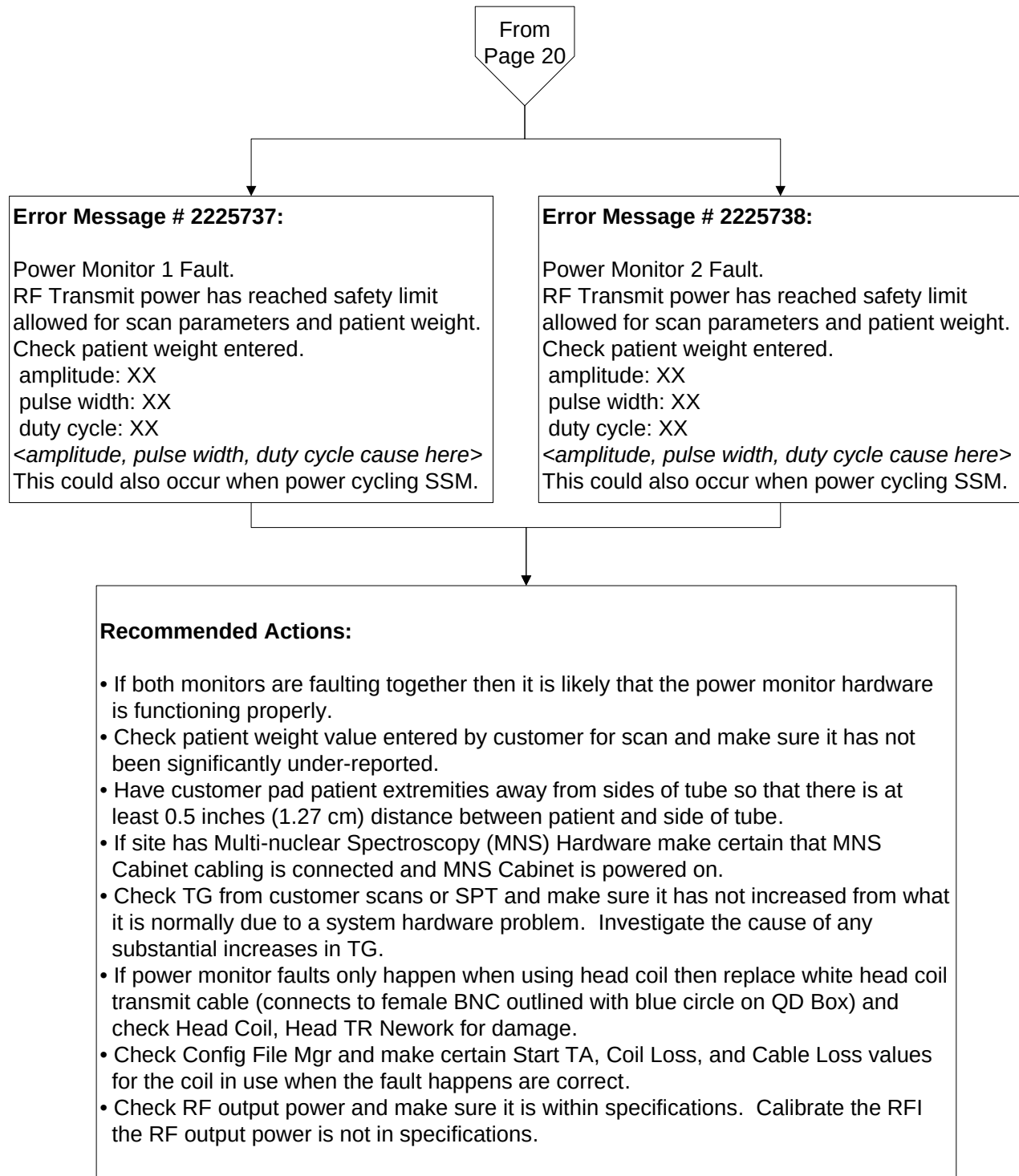
Note: Under normal conditions you should not see this error code. The SSM no longer performs these functions.



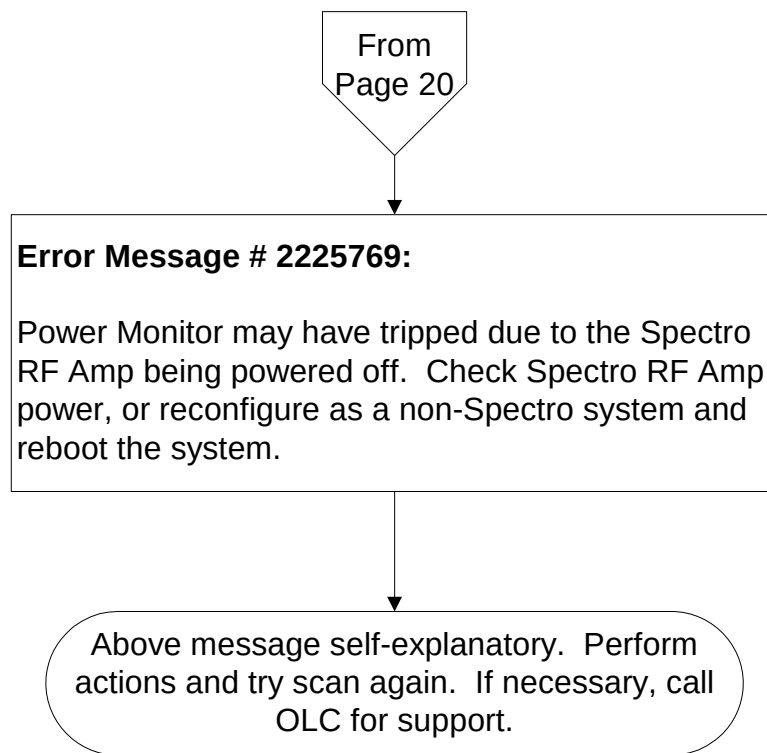
2-41 Error Message # 2225570



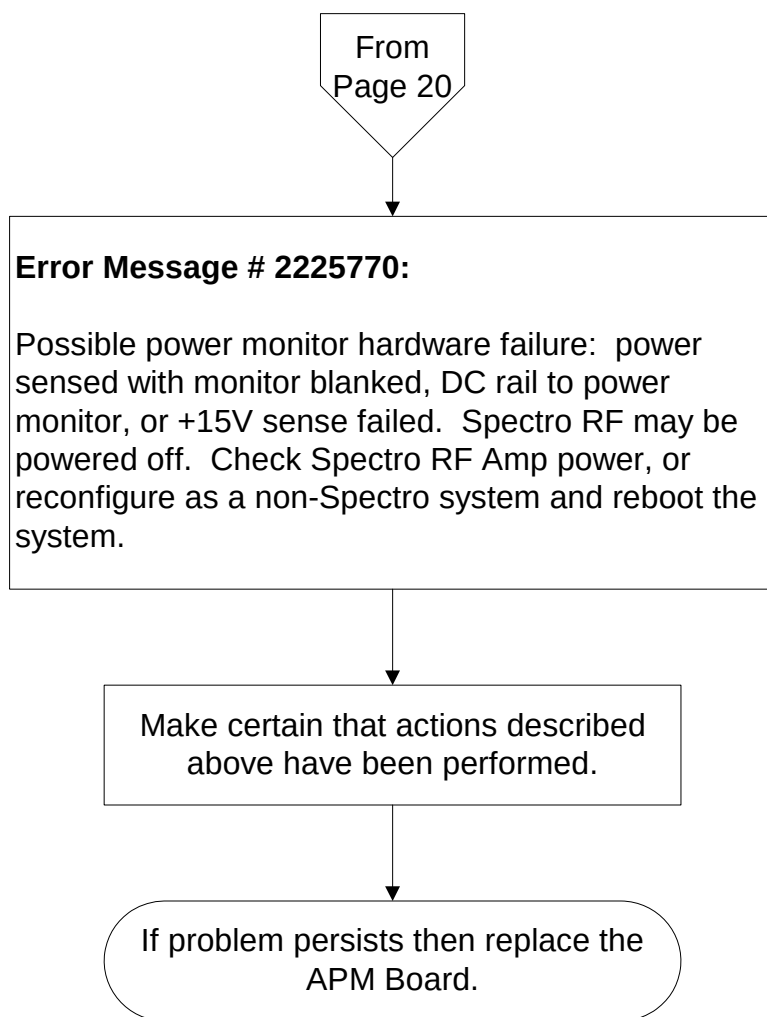
2-42 Error Messages 2225737 & 2225738



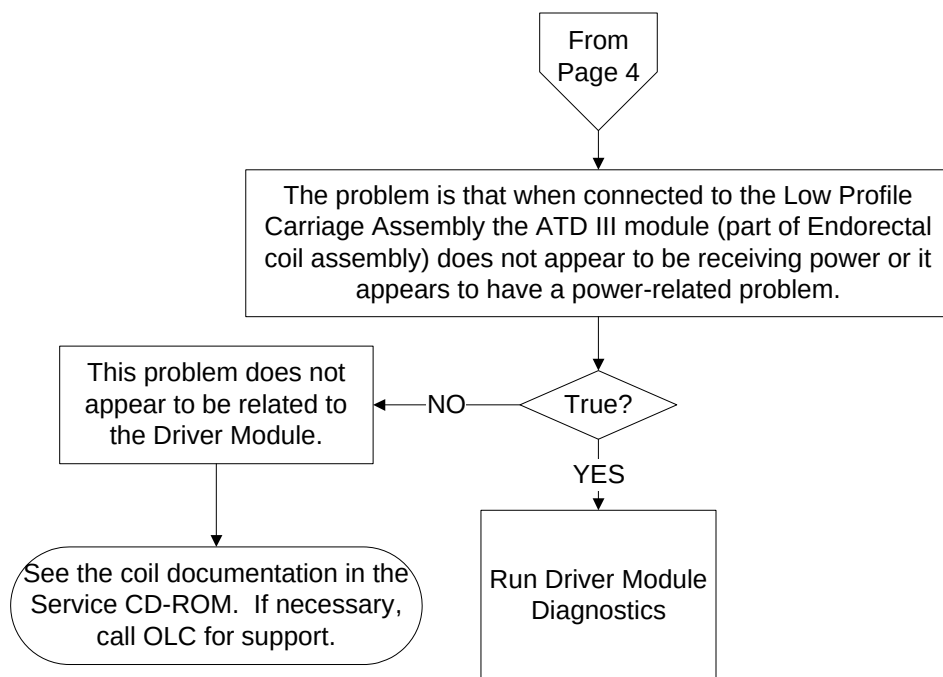
2-43 Error Message # 2225769



2-44 Error Message # 2225770



2-45 ATD III Troubleshooting



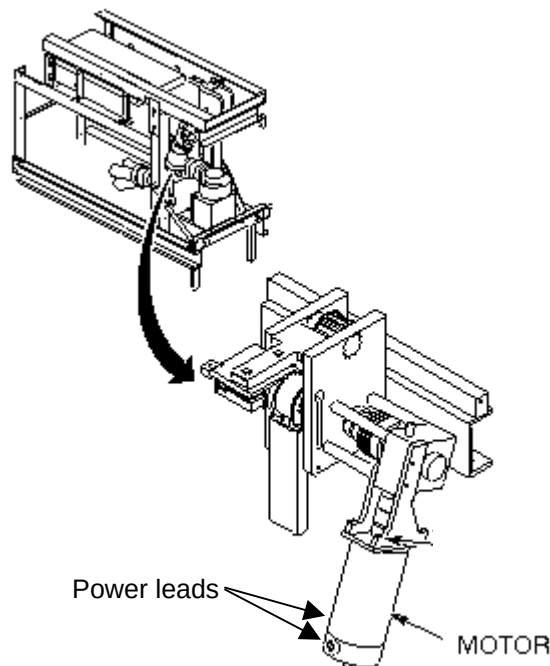
3 – TROUBLESHOOTING LONGITUDINAL DRIVE FAILURES

This section is divided into two parts. **Section 3-1 Troubleshooting Lack of Carriage Motion** should be performed if, when the FAST IN buttons on the Magnet Enclosure are pressed, the Longitudinal Drive Motor cannot be heard rotating and, as a result, the Carriage does not move. If the problem was not isolated in the previous section or if the motor rotates but carriage motion is erratic or poor then also perform the actions in **Section 3-2 Troubleshooting Carriage Motion Problems**. This section will assist the FE with determining if carriage motion / longitudinal drive motor rotation problems are due to components within the SSM. This process requires two Field Engineers. The FEs may also want to reference the System Support Module block diagrams located in the TPS/ISE tab of **Signa Release 12.X or 14.X Block Diagrams & Supplemental Schematics** from the Service CD-ROM during this process.

3-1 Troubleshooting Lack of Carriage Motion

Perform the steps in this section if the Longitudinal Drive Motor cannot be heard rotating when the FAST IN buttons on the Magnet Enclosure are pressed and, as a result, the Carriage does not move. Skip this section if the Carriage moves but the motion is poor. If in doubt, perform both sections.

1. Remove the right cover from the rear pedestal to expose the longitudinal drive motor. See Illustration 3-1.



RIGHT COVER REMOVED TO EXPOSE LONGITUDINAL DRIVE MOTOR
ILLUSTRATION 3-1

1. Confirm that the two spade lead power connectors are firmly attached to the spade lugs on either side of the longitudinal motor. See Illustration 3-1.

3-1 Troubleshooting Lack of Carriage Motion (Continued)

2. Verify that the sub-D connector labeled Run 318 J11 is firmly connected to J11 at the magnet room side of the penetration panel.
3. Verify that the sub-D connector labeled Run 770 J11 is firmly connected to J11 at the equipment room side of the penetration panel.
4. Verify that the sub-D connector labeled Run 770 J41 is firmly connected to the **J41 LONG AMP** connector on the rear of the SSM.
5. Locate the SRI Module.
6. Locate LED **DS31** (UPLIM). Verify that with the table in the full-up position this LED is illuminated.

3-1 Troubleshooting Lack of Carriage Motion (Continued)

NOTE

The DOCKSTP signal, which corresponds to LED **DS8**, is no longer monitored by the Horizon LX SRI.

7. If LED **DS9** is not illuminated:
 - a. Check the cable connections on either side of Run 421 between J1 on the Dock Limit Switch Box and J3 on the SRI.
 - b. Check cable connection between cable from Dock Assembly and J2 on the Dock Limit Switch Box.



FERROUS MATERIAL HAZARD!! THE DOCK ASSEMBLY CONTAINS A MOTOR MADE WITH FERROUS COMPONENTS. DO NOT DISASSEMBLE THE DOCK ASSEMBLY INSIDE THE MAGNET ROOM. INSTEAD, REFER TO SECTION 3 – DOCK COMPONENTS IN THE SRI, DOCK HARDWARE (ME1REA1.DOC) DOCUMENT AND REMOVE THE DOCK FROM THE ROOM BEFORE DISASSEMBLY AS DESCRIBED IN THE PROCEDURE.

- c. Check the wiring to and the functionality of the Up-Limit Switch inside the dock. This switch contacts with a plunger on the front of the table and allows the system to tell when the table is in the full-up position.
8. Remove the cover from the Dock Limit Switch Box to expose the Dock Limit Switch Board.
9. Locate LED DS15 (8 VDC supply) in the upper-left side of the Dock Limit Switch Board and confirm it is illuminated. This LED will be illuminated if the Dock Limit Switch Board is receiving 8 VDC power from the SRI.

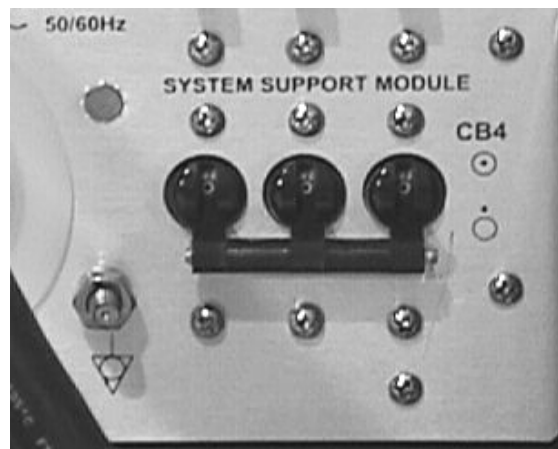
3-2 Troubleshooting Carriage Motion Problems

Perform the steps in this section if the steps in **Section 3-1 Troubleshooting Lack of Carriage Motion** were already performed or if the Carriage moves but has poor motion. This section assumes that the FE has already verified that the Drive Belt was correctly routed and that the Longitudinal Motor Drive hardware in the Rear Pedestal is correctly adjusted and functioning properly.

1. Locate the **CB4 SYSTEM SUPPORT MODULE** Circuit Breaker in the lower-right corner rear of the SSM and place it in the down (OFF) position. See Illustration 3-2.



Placing the CB4 System support module circuit breaker in the down (OFF) position will remove power from high voltage Dynamic Disable and Inductive Drive Disable Circuits but does not remove 120 VAC power from the RF Amplifiers, Spectroscopy RF Amplifier, Front Service Outlets, or Gradient Blower Fan.

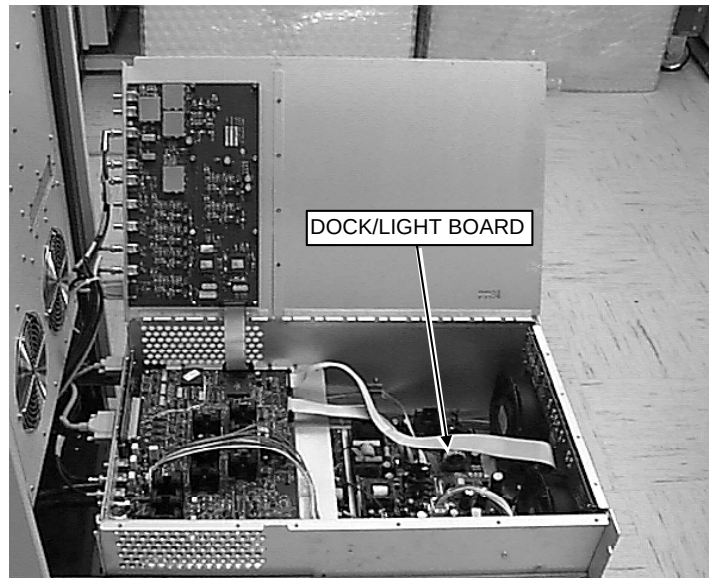


CB4 SYSTEM SUPPORT MODULE CIRCUIT BREAKER IN DOWN (OFF) POSITION
ILLUSTRATION 3-2

1. Refer to **APPENDIX A – REMOVING AND REPLACING SSM CHASSIS LID** and remove SSM lid.
2. Obtain Digital Multimeter (DMM) with clip-on test leads and configure to measure at least 50 VDC.

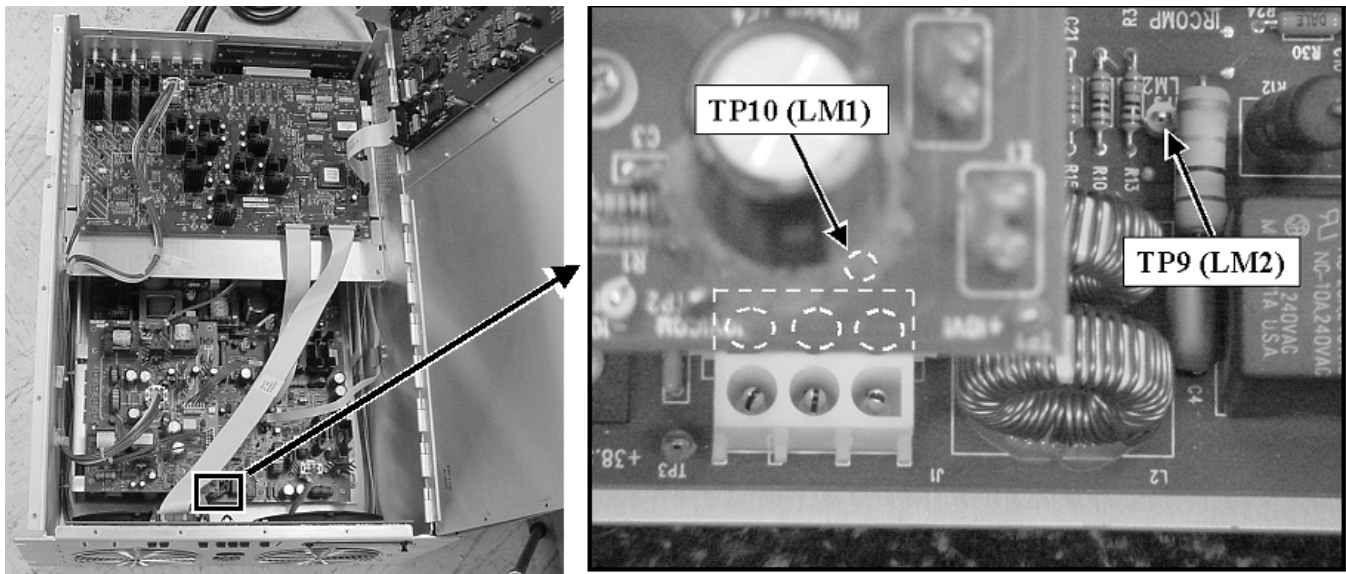
3-2 Troubleshooting Carriage Motion Problems (Continued)

3. Refer to Illustration 3-3 and locate Dock/Light Board inside bottom of SSM. This board is mounted on the bottom of the SSM chassis in front of the fans.



LOCATION OF DOCK/LIGHT BOARD INSIDE SSM
ILLUSTRATION 3-3

4. Clip DMM test leads to **TP9 (LM2)** and **TP10 (LM1)** test points on front of Dock/Light Board. See Illustration 3-4. Notice that TP10 (LM1) is located underneath the HV Board. Lead polarity is not important. The voltage measured between **TP9 (LongMtr2)** and **TP10 (LongMtr1)** is the Longitudinal Drive Motor voltage.



LOCATION OF TP9 AND TP10 TEST POINTS ON DOCK/LIGHT BOARD
ILLUSTRATION 3-4

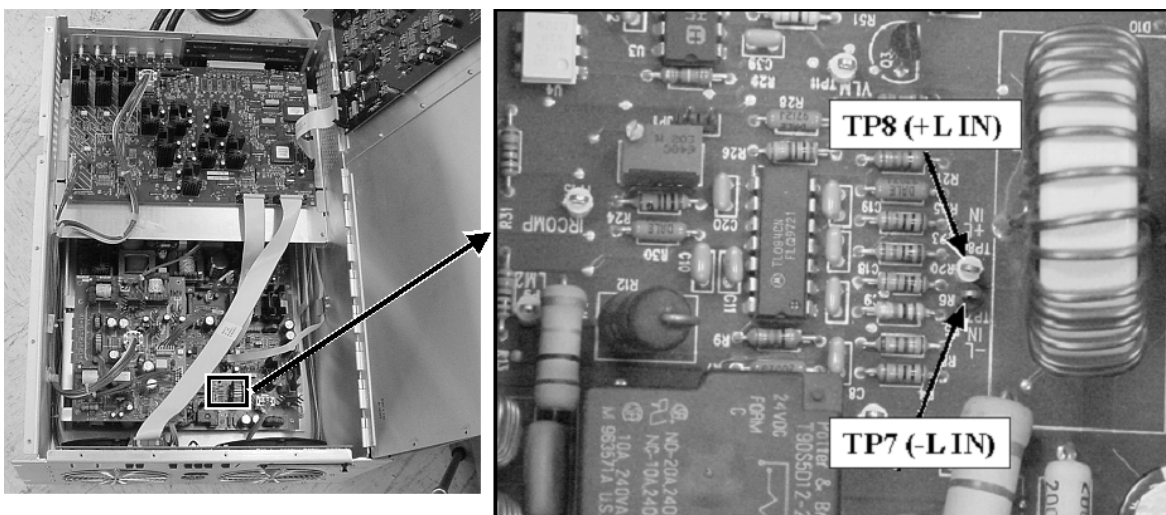
5. Close SSM lid and let it rest on test leads.

3-2 Troubleshooting Carriage Motion Problems (Continued)

WARNING!

DO NOT LEAVE THE SSM LID OPEN IN HEAD OR SURFACE MODE WITH POWER ON. THE SSM WILL OVERHEAT AND COMPONENT DAMAGE WILL RESULT. IN HEAD OR SURFACE MODE THE LID MUST BE LEFT IN THE CLOSED POSITION WHEN POWER IS APPLIED TO INSURE ADEQUATE AIR FLOW THROUGH THE SSM.

6. Apply power to the SSM.
7. Have someone in the Magnet Room push the IN FAST button on the magnet enclosure. Watch the DMM display and note the voltage. In normal operation this voltage should range between 17 and 20 VDC. If the cradle isn't moving then this voltage should ramp up to about 30 volts and then, then the system determines that there is no movement, it will ramp down to 0 (zero) volts.
8. **If the voltage is not present or is very low:**
 - a. Remove the cable connected to **J41 LONG AMP** on the rear of the SSM and set it aside.
 - b. Measure the voltage again while driving the carriage into the bore.
9. **If the voltage across TP9 and TP10 is still not present:**
 - c. Carefully remove test leads from **TP9** and **TP10** and connect them to **TP7 (-L IN)** and **TP8 (+L IN)** on the Dock/Light Board. See Illustration 3-5. Again, lead polarity is not important. Test points **TP7** and **TP8** allow convenient measurement of the MOTORCMD signal from the SRI that drives the Longitudinal Drive Motor.



LOCATION OF TP7 AND TP8 ON FRONT OF DOCK / LIGHT BOARD
ILLUSTRATION 3-5

3-2 Troubleshooting Carriage Motion Problems (Continued)

- d. While attempting to drive the carriage into the bore verify that the voltage on the DMM display “spikes” up to 9VDC and then to 0 (zero) VDC. In normal operation this voltage “ramps” up to a steady 6 VDC when the carriage starts moving and then, when the carriage movement ends, the voltage “ramps” down from a steady 6 VDC to 0 (zero) VDC.
 - i. **If the voltage across TP7 and TP8 is not seen or is seen but is very low:**
 - ii. Check for poor cable connections from **J40 SRI POWER** on the rear of the SSM to **J14** on the Penetration Panel to **J1** on the SRI .
 - iii. If these all appear OK then replace the SRI.
 - iv. **If the voltage is seen across TP7 and TP8** then replace the Dock/Light Board.
- 10. **If the voltage across TP9 and TP10 is present but low** then replace the Dock/Light Board.
- 11. **If the voltage across TP9 and TP10 is present and in range** then either the K1 relay isn't working or one of the circuit load (longitudinal drive motor, pen panel filters, cable connectors, cables) components are possibly open-circuited.



THE MOTOR IS MADE OUT OF FERROUS COMPONENTS AND WILL BE ATTRACTED TO THE MAGNET IF REMOVED FROM THE REAR PEDESTAL ASSEMBLY. REFER TO SECTION 3 – LONGITUDINAL DRIVE MOTOR REPLACEMENT IN THE APPROPRIATE REAR PEDESTAL HARDWARE REPLACEMENT PROCEDURE (CX OR NON-CX PROCEDURE) BEFORE ATTEMPTING TO REMOVE THE MOTOR FROM THE REAR PEDESTAL.

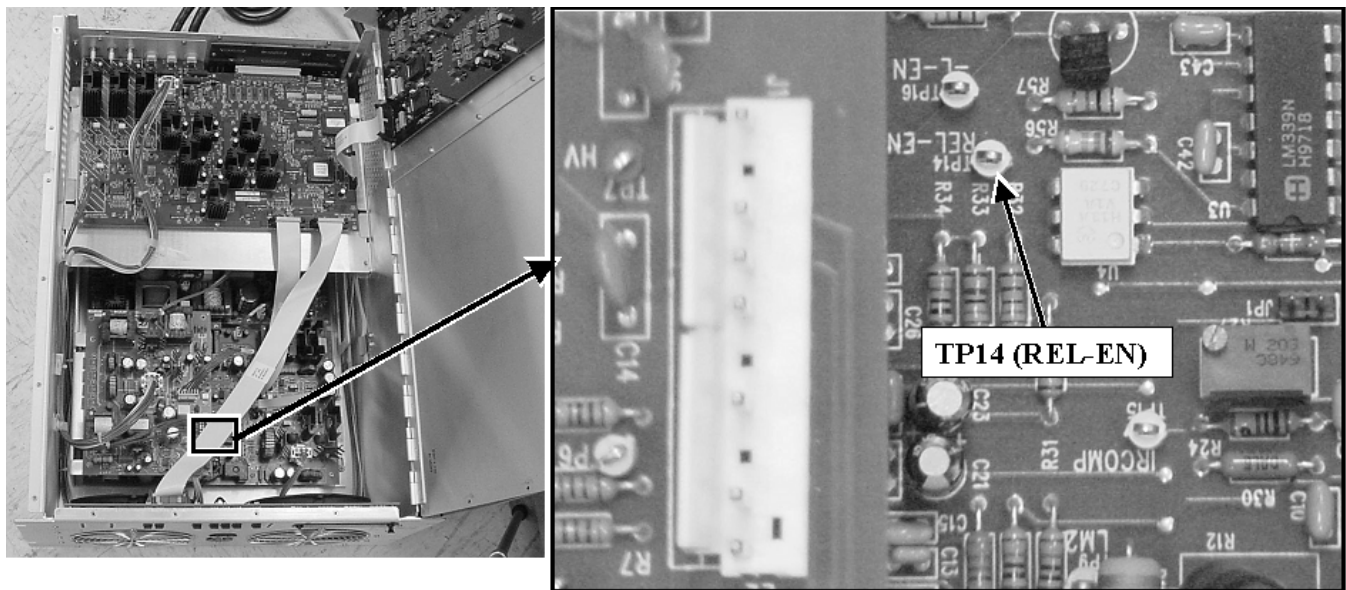
- a. The longitudinal drive motor can be tested outside the magnet room by removing it from the rear pedestal and then connecting it to the cable that is normally connected to **J11** (RUN 770) on the equipment room side of the penetration panel. See the **DANGER** message above and refer to documented removal process to remove the motor from the rear pedestal.
- b. Have someone in the Magnet Room push the IN FAST button on the magnet enclosure. Measure the voltage across the spade lug connections on the motor.
 - i. If the voltage ramps up to 30 volts but the motor is still not rotating then check the motor brushes and, if necessary, replace the motor.
 - ii. If the motor now works then inspect the penetration panel filter and sub-D connectors at **J11** on the Equipment and Magnet Room sides of the Penetration Panel for corrosion or obvious signs of damage.

3-2 Troubleshooting Carriage Motion Problems (Continued)

WARNING!

DO NOT LEAVE THE SSM LID OPEN IN HEAD OR SURFACE MODE WITH POWER ON. THE SSM WILL OVERHEAT AND COMPONENT DAMAGE WILL RESULT. IN HEAD OR SURFACE MODE THE LID MUST BE LEFT IN THE CLOSED POSITION WHEN POWER IS APPLIED TO INSURE ADEQUATE AIR FLOW THROUGH THE SSM.

- c. If the 30 volt ramp is still not seen when someone is pressing the IN FAST button then open the SSM top cover and locate the **TP14 (REL-EN)** on the Dock / Light Board. See Illustration 3-6. Connect the DMM test leads from **TP14 (REL-EN)** to chassis ground. Polarity is not important.



LOCATION OF TP14 ON FRONT OF DOCK / LIGHT BOARD
ILLUSTRATION 3-6

- d. Have someone press one of the FAST IN buttons on either Magnet Enclosure Keypad. The voltage should read about 30.9 volts when the IN FAST button is not pressed and should change to 10.9 volts when the IN FAST button is pressed.
 - i. **If no voltage change was seen** then the Out Enable signal from SRI may not be reaching SSM. Check the continuity of the line between the SRI and the SSM. Pins 3 and 4 on the J7 cable carry the differential signals between the SRI and SSM. Otherwise, replace the SRI.
 - ii. **If change was seen** then Dock/Light Board may have failed. Replace Dock/Light Board.

3-2 Troubleshooting Carriage Motion Problems (Continued)

12. Remove the test leads and perform actions in **Section A-2 Replacing SSM Chassis Lid**.
13. Perform one head and body scan, if possible.
14. Return to troubleshooting flowchart, if necessary.

4 – TROUBLESHOOTING BORE FAN FAILURE

This section is divided into two parts. **Section 4-1 Checking Motor AC Voltage** provides the FE with a process for determining that the motor is receiving power. **Section 4-2 Checking Functionality of SSM Bore Fan Switching Circuitry** provides the FE with a process for determining that the failure is not within the SSM.

4-1 Checking Motor AC Voltage

1. Confirm that the Gradient Fan runs normally.
2. Refer to Illustration 4-1.
 - a. Place the **CB1 BORE VENT FAN** Circuit Breaker in the down (OFF) position.
 - b. Place the **CB2 GRADIENT FAN** Circuit Breaker in the down (OFF) position.
 - c. Remove the cable connected to **J37 BORE VENT FAN 300VA** and set this cable aside.
 - d. Remove the cable connected to **J47 GRAD. FAN 900VA** and set this cable aside.



BORE VENT CB1 CIRCUIT BREAKER LOCATION
ILLUSTRATION 4-1

3. Locate the Bore Vent cable labeled MR1-A7-J37 removed in the last step and connect this cable instead to the **J47 GRAD. FAN 900VA** connector on the rear of the SSM.
4. Place the **CB2 GRADIENT FAN** Circuit Breaker in the up (ON) position.
5. Confirm that the bore vent fan is running normally.
6. **If the fan is not running normally** then place the **CB2 GRADIENT FAN** Circuit Breaker in the down (OFF) position.
 - a. Remove the Bore Vent Cable labeled MR1-A7-J37 from the **J47 GRAD. FAN 900VA** connector on the rear of the SSM.
 - b. Re-connect the Bore Vent Cable labeled MR1-A7-J37 to the **J37 BORE VENT FAN 300VA** connector on the rear of the SSM.

4-1 Checking Motor AC Voltage (Continued)

- c. Re-connect the Gradient Fan Cable labeled MR1-A7-J47 to the **J47 GRAD. FAN 900VA** connector on the rear of the SSM.
- d. Refer to the **BLOWER BOX MOTOR REPLACEMENT** procedure (MI0REA2.DOC) and replace the bore fan motor.
- e. After finished with the Bore Fan Motor replacement, place the **CB1 BORE FAN VENT 300VA** Circuit Breaker on the rear of the SSM in the up (ON) position. Confirm that the Bore Fan is running.
- f. Return, if necessary, to the troubleshooting flowchart.

7. **If the fan is running normally** then proceed to the next section.

4-2 Checking Functionality of SSM Bore Fan Switching Circuitry

This section assumes that all the checks described in **Section 4-1 Checking Motor AC Voltage** have been completed and that the Bore Fan Motor is known to be functional.

1. Perform Button Test Diagnostic. Refer to Functional Checks/Magnet Enclosure/Magnet Enclosure Button Test.
2. Wait about 10 seconds and then enter the Magnet Room.
 - a. Press the buttons on the right and left Magnet Enclosure Keypads that toggle power to the Bore Fan Motor on and off.
 - b. Confirm that each time a button is pressed a unique number and a number showing the total of number of times a button was pressed are displayed in the Scan Time and Table Motion Display Windows, respectively.

4-2 Checking Functionality of SSM Bore Fan Switching Circuitry (Continued)

3. If one or both of the buttons don't respond then check the functionality of the SRI and the keypad(s).
4. If the test was successful then cancel out of the test and exit the diagnostics.
5. Completely remove power from the SSM by placing the System Support Module circuit breaker on the front of the PDU in the OFF position and then Locking and Tagging out the panel per Safety requirements.

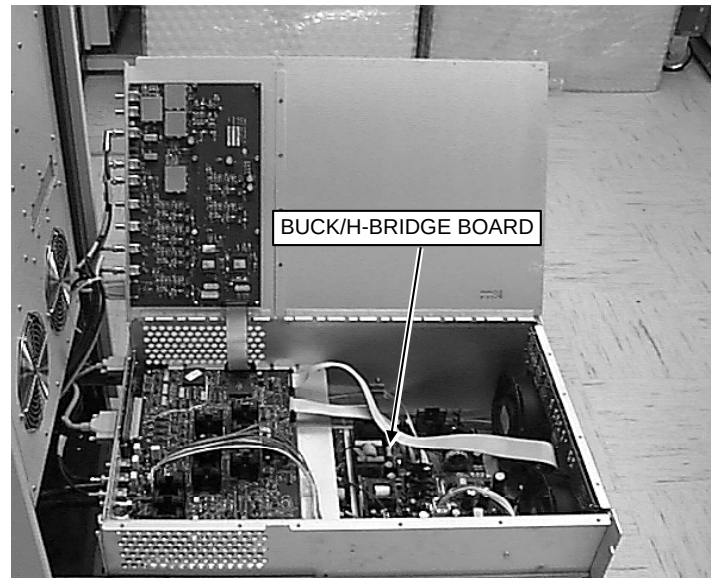


AVOID SERIOUS INJURY OR DEATH BY ELECTROCUTION. LOCK AND TAGOUT THE SYSTEM SUPPORT MODULE CIRCUIT BREAKER AT THE PDU AS THE SSM CIRCUITS TO BE CHECKED NEXT WILL NOT BE DE-ENERGIZED BY PLACING ANY OF THE CIRCUIT BREAKERS ON THE REAR OF THE SSM IN THE OFF POSITION.

6. Refer to **APPENDIX A – REMOVING AND REPLACING SSM CHASSIS LID** and remove SSM lid.
7. Obtain Digital Multimeter (DMM) with test leads.
8. Wait 5 minutes for the SSM power supplies to discharge before opening the SSM lid.

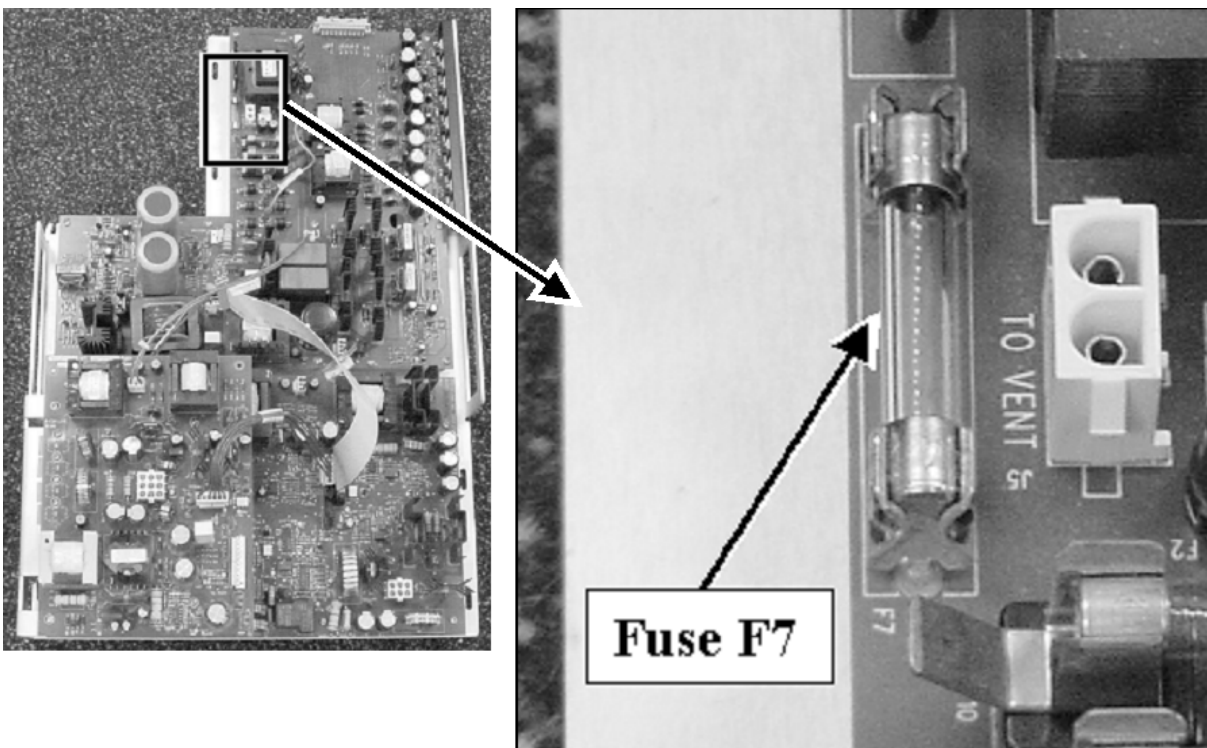
4-2 Checking Functionality of SSM Bore Fan Switching Circuitry (Continued)

9. Refer to Illustration 4-2 and locate Buck/H-Bridge Board inside bottom of the SSM. This board is mounted on the bottom of the SSM chassis behind the Dock/Light Board.



LOCATION OF BUCK/H-BRIDGE BOARD INSIDE SSM
ILLUSTRATION 4-2

10. Remove the aluminum plate that the CPD Board is affixed so as to gain access to fuse F7 on the Buck/H-Bridge Board. See Illustration 4-3.



FUSE F7 LOCATED ON REAR OF BUCK / H-BRIDGE BOARD
ILLUSTRATION 4-3

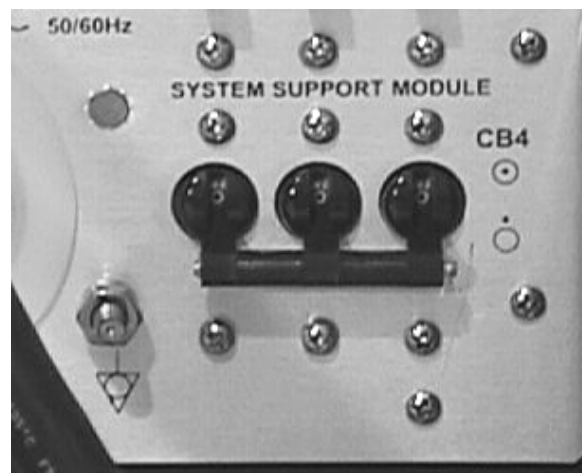
4-2 Checking Functionality of SSM Bore Fan Switching Circuitry (Continued)

11. See Illustration 4-5 and locate 4 amp fuse F7 on the Buck/H-Bridge Board. Use the DMM to check the continuity of the fuse and verify that it is not open.
12. Verify the electrical connection between CB4 SYSTEM SUPPORT MODULE Circuit Breaker and the white J5 connector seen next to fuse F7 in Illustration 4-6 on the Buck/H-Bridge Board.
13. Use the DMM to check the continuity of each leg of the CB1 circuit breaker.
14. If the fuse is open and the Bore Fan is known to be in good working order then replace either the Buck / H-Bridge Board or the SSM.
15. If the CB1 circuit breaker is found to be faulty then replace either the circuit breaker or the SSM.
16. When finished, confirm that all the test leads have been removed and all the SSM cabling has been reconnected.
17. Re-install the CPD Board assembly into the SSM and then close the SSM lid.
18. Remove the Lockout / Tagout hardware from the System Support Module circuit breaker on the front of the PDU and restore power to the SSM.
19. Perform actions in **Section A-2 Replacing SSM Chassis Lid**.
20. Perform one head and body scan, if possible.
21. Return to troubleshooting flowchart, if necessary.

5 – TROUBLESHOOTING BORE LIGHT FAILURE

This section assumes that the FE has already checked the halogen lamp filament and confirmed that the lamp is in good working order.

1. Perform Button Test Diagnostic. Refer to Functional Checks/Magnet Enclosure/Magnet Enclosure Button Test.
 - a. Press the buttons on the right and left Magnet Enclosure Keypads that toggle power to the Bore Light on and off.
 - b. Confirm that each time a button is pressed a unique number and a number showing the total of number of times a button was pressed are displayed in the Scan Time and Table Motion Display Windows, respectively.
2. If one or both of the buttons don't respond then check the functionality of the SRI and the keypad(s).
3. If the test was successful then cancel out of the test and exit the diagnostics.
4. Locate the **CB4 SYSTEM SUPPORT MODULE** Circuit Breaker in the lower-right corner rear of the SSM and place it in the down (OFF) position. See Illustration 5-1.



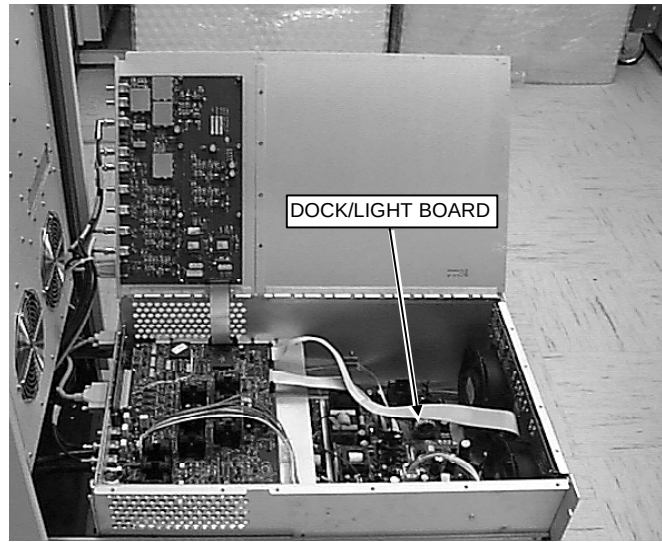
CB4 SYSTEM SUPPORT MODULE CIRCUIT BREAKER IN DOWN (OFF) POSITION
ILLUSTRATION 5-1



Placing the CB4 System support module circuit breaker in the down (OFF) position will remove power from high voltage Dynamic Disable and Inductive Drive Disable Circuits but does not remove AC power from the NB AMP receptacle, SPECTRO AMP receptacle, Front Service Outlets, or Gradient Blower Fan.

5 – TROUBLESHOOTING BORE LIGHT FAILURE (CONTINUED)

5. Refer to **APPENDIX A – REMOVING AND REPLACING SSM CHASSIS LID** and remove SSM lid.
6. Obtain Digital Multimeter (DMM) with clip-on test leads and configure to measure 5VDC.
7. Refer to Illustration 5-2 and locate Dock/Light Board inside bottom of the SSM. This board is mounted on the bottom of the SSM chassis behind the fans.

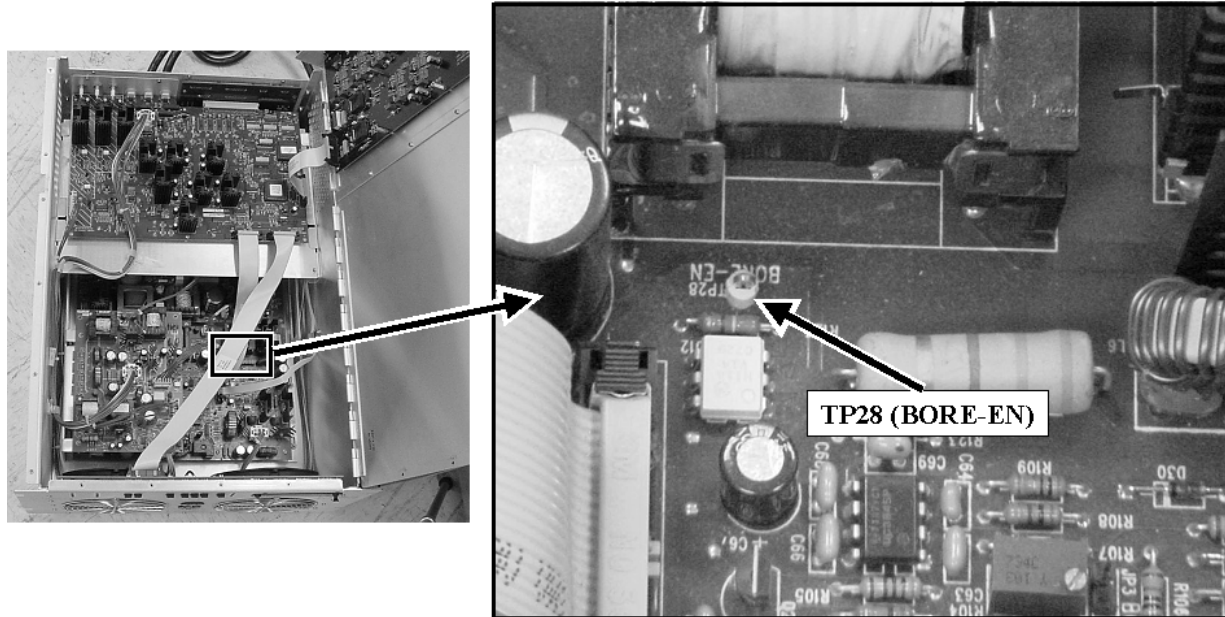


LOCATION OF DOCK/LIGHT BOARD INSIDE SSM
ILLUSTRATION 5-2

8. Reseat the ribbon cable interconnecting connector J4 on the Dock/Light Board to J9 on the CPD Board.

5 – TROUBLESHOOTING BORE LIGHT FAILURE (CONTINUED)

9. Refer to Illustration 5-3 and locate **TP28 (BORE-EN)** on the Dock/Light Board. Connect one DMM test lead to TP28 and the other side to chassis ground. Lead polarity is not important.



LOCATION OF TP28 ON THE DOCK/LIGHT BOARD
ILLUSTRATION 5-3

5 – TROUBLESHOOTING BORE LIGHT FAILURE (CONTINUED)

10. Connect other DMM test lead to SSM chassis ground.
11. Close SSM lid and let it rest on test leads.
12. Locate the **CB4 SYSTEM SUPPORT MODULE** Circuit Breaker in the lower-right corner rear of the SSM and place it in the up (ON) position.

WARNING!

DO NOT LEAVE THE SSM LID OPEN IN HEAD OR SURFACE MODE WITH POWER ON. THE SSM WILL OVERHEAT AND COMPONENT DAMAGE WILL RESULT. IN HEAD OR SURFACE MODE THE LID MUST BE LEFT IN THE CLOSED POSITION WHEN POWER IS APPLIED TO INSURE ADEQUATE AIR FLOW THROUGH THE SSM.

5 – TROUBLESHOOTING BORE LIGHT FAILURE (CONTINUED)

13. Have someone push one of the Bore Light Buttons on either of the keypads on the front of the Magnet Enclosure in the Magnet Room. The DMM should indicate a 5 volt spike.
 - a. **If the voltage is 0 (zero) VDC or close to this** then replace the CPD Board.
 - b. **If the voltage is 5 VDC or close to this** then the signal is reaching the SSM. Replace the Dock/Light Board.
14. Power off the SSM.
15. Carefully remove the DMM test leads from inside the SSM.
16. Perform actions in **Section A-2 Replacing SSM Chassis Lid**.
17. Perform one head and body scan, if possible.
18. If necessary, return to the troubleshooting flowchart.

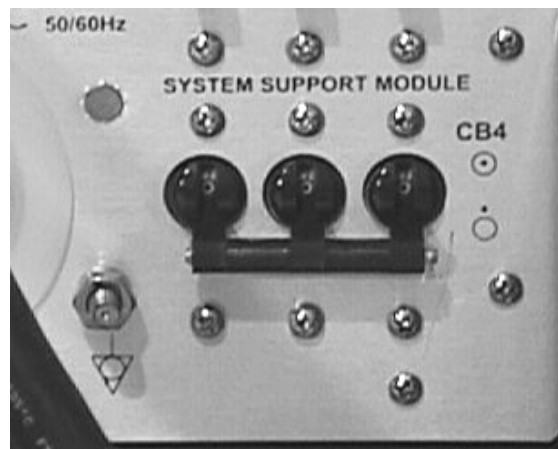
6 – TROUBLESHOOTING PATIENT ALIGNMENT LIGHT FAILURE

This section assumes that the FE has already checked the halogen lamp filament and confirmed that the lamp is in good working order.

1. Perform Button Test Diagnostic. Refer to Functional Checks/Magnet Enclosure/Magnet Enclosure Button Test.
 - a. Press the buttons on the right and left Magnet Enclosure Keypads that toggle power to the Patient Alignment Light on and off.
 - b. Confirm that each time a button is pressed a unique number and a number showing the total of number of times a button was pressed are displayed in the Scan Time and Table Motion Display Windows, respectively.
2. If one or both of the buttons don't respond then check the functionality of the SRI and the keypad(s).
3. If the test was successful then cancel out of the test and exit the diagnostics.
4. Locate the **CB4 SYSTEM SUPPORT MODULE** Circuit Breaker in the lower-right corner rear of the SSM and place it in the down (OFF) position. See Illustration 6-1.

CAUTION

Placing the CB4 System support module circuit breaker in the down (OFF) position will remove power from high voltage Dynamic Disable and Inductive Drive Disable Circuits but does not remove AC power from the NB AMP receptacle, SPECTRO AMP receptacle, Front Service Outlets, or Gradient Blower Fan.

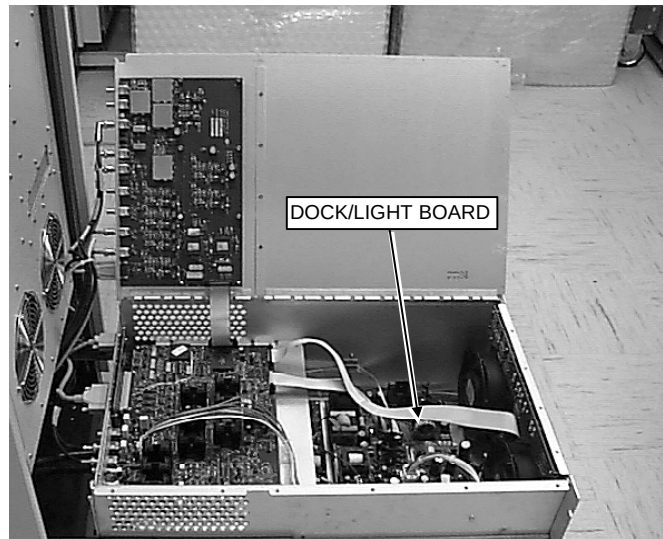


CB4 SYSTEM SUPPORT MODULE CIRCUIT BREAKER IN DOWN (OFF) POSITION
ILLUSTRATION 6-1

5. Refer to **APPENDIX A – REMOVING AND REPLACING SSM CHASSIS LID** and remove SSM lid.
6. Obtain Digital Multimeter (DMM) with clip-on test leads and configure to measure 5VDC.

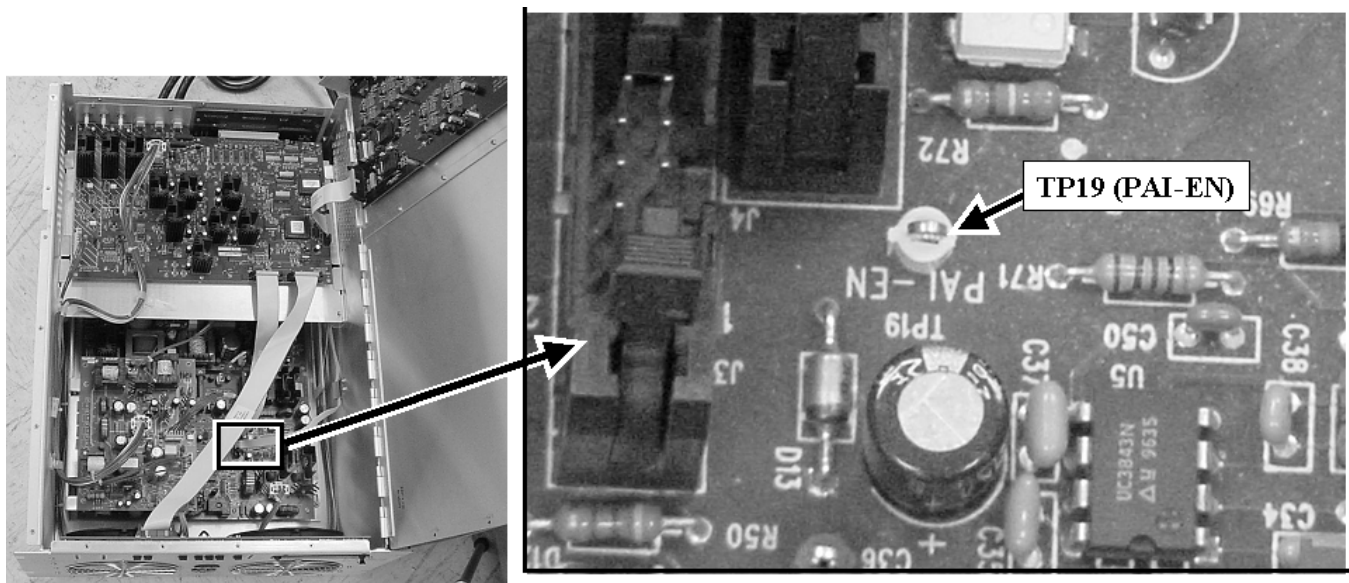
6 – TROUBLESHOOTING PATIENT ALIGNMENT LIGHT FAILURE (CONTINUED)

7. Refer to Illustration 6-2 and locate Dock/Light Board inside bottom of the SSM. This board is mounted on the bottom of the SSM chassis behind the fans.



LOCATION OF DOCK/LIGHT BOARD INSIDE SSM
ILLUSTRATION 6-2

8. Reseat the ribbon cable interconnecting connector J4 on the Dock/Light Board to J9 on the CPD Board.
9. Refer to Illustration 6-3 and locate **TP19 (PAL-EN)** on the Dock/Light Board. Connect one DMM test lead to TP19 and the other to chassis ground.



LOCATION OF TP19 ON THE DOCK / LIGHT BOARD
ILLUSTRATION 6-3

10. Locate the **CB4 SYSTEM SUPPORT MODULE** Circuit Breaker in the lower-right corner rear of the SSM and place it in the up (ON) position.

6 – TROUBLESHOOTING PATIENT ALIGNMENT LIGHT FAILURE (CONTINUED)

WARNING!

DO NOT LEAVE THE SSM LID OPEN IN HEAD OR SURFACE MODE WITH POWER ON. THE SSM WILL OVERHEAT AND COMPONENT DAMAGE WILL RESULT. IN HEAD OR SURFACE MODE THE LID MUST BE LEFT IN THE CLOSED POSITION WHEN POWER IS APPLIED TO INSURE ADEQUATE AIR FLOW THROUGH THE SSM.

11. Have someone push one of the Patient Alignment Light Buttons on either of the keypads on the front of the Magnet Enclosure in the Magnet Room.
12. Check the DMM display for any change in voltage level. If the optocoupler is active then the voltage shown on the DMM display should spike up to 5 volts, then down.
 - a. **If the voltage is 0 (zero) VDC or close to this** then replace the CPD Board.
 - b. **If the voltage is 5 VDC or close to this** then the signal has reached the optocoupler. Replace the Dock/Light Board.
13. Power off the SSM.
14. Carefully remove the DMM test leads from inside the SSM.
15. Perform actions in **Section A-2 Replacing SSM Chassis Lid**.
16. Perform one head and body scan, if possible.
17. If necessary, return to the troubleshooting flowchart.

7 – TROUBLESHOOTING SSM RELATED IMAGE QUALITY PROBLEMS

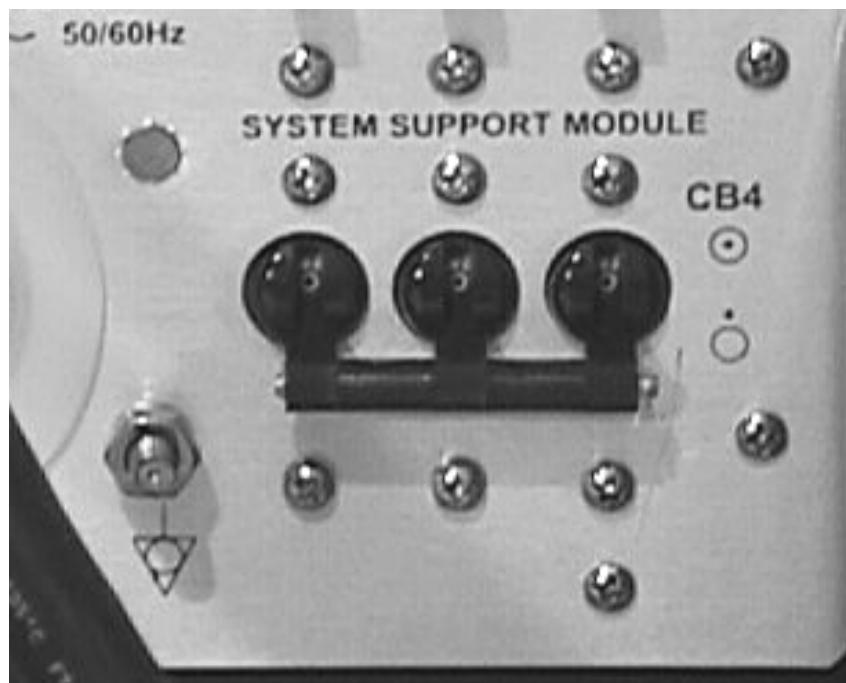
7-1 Poor Image Quality Due To Poor DD Bias Line Connections

Note: Under normal conditions you should not see these error codes. The SSM no longer performs these functions.

7-2 Poor Image Quality Due to Loss of High Voltage Bias

Note: HV on the SSM is not used in the system. HV is provided by the Driver Module.

1. Locate the **CB4 SYSTEM SUPPORT MODULE** Circuit Breaker in the lower-right corner rear of the SSM and place it in the down (OFF) position. See Illustration 7-1.



CB4 SYSTEM SUPPORT MODULE CIRCUIT BREAKER IN DOWN (OFF) POSITION
ILLUSTRATION 7-1

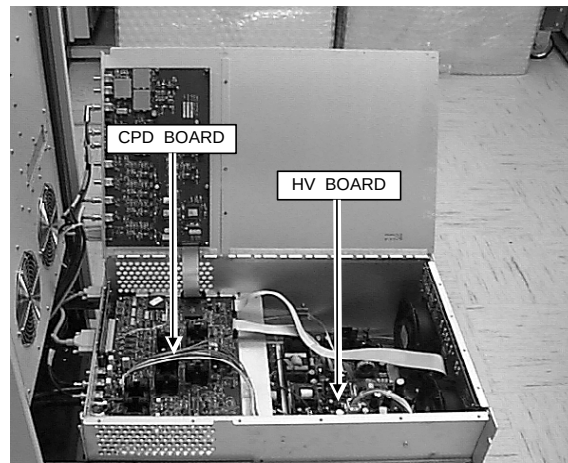
Note: The Driver Module now provides all head, body, and multicoil TR biasing functions as well as all dynamic disable and direct drive biasing.



Placing the CB4 System support module circuit breaker in the down (OFF) position will remove power from high voltage Dynamic Disable and Inductive Drive Disable Circuits but does not remove 120 VAC power from the Erbtac RF Amplifier, Spectroscopy RF Amplifier, Front Service Outlets, or Gradient Blower Fan.

7-2 Poor Image Quality Due to Loss of High Voltage Bias (Continued)

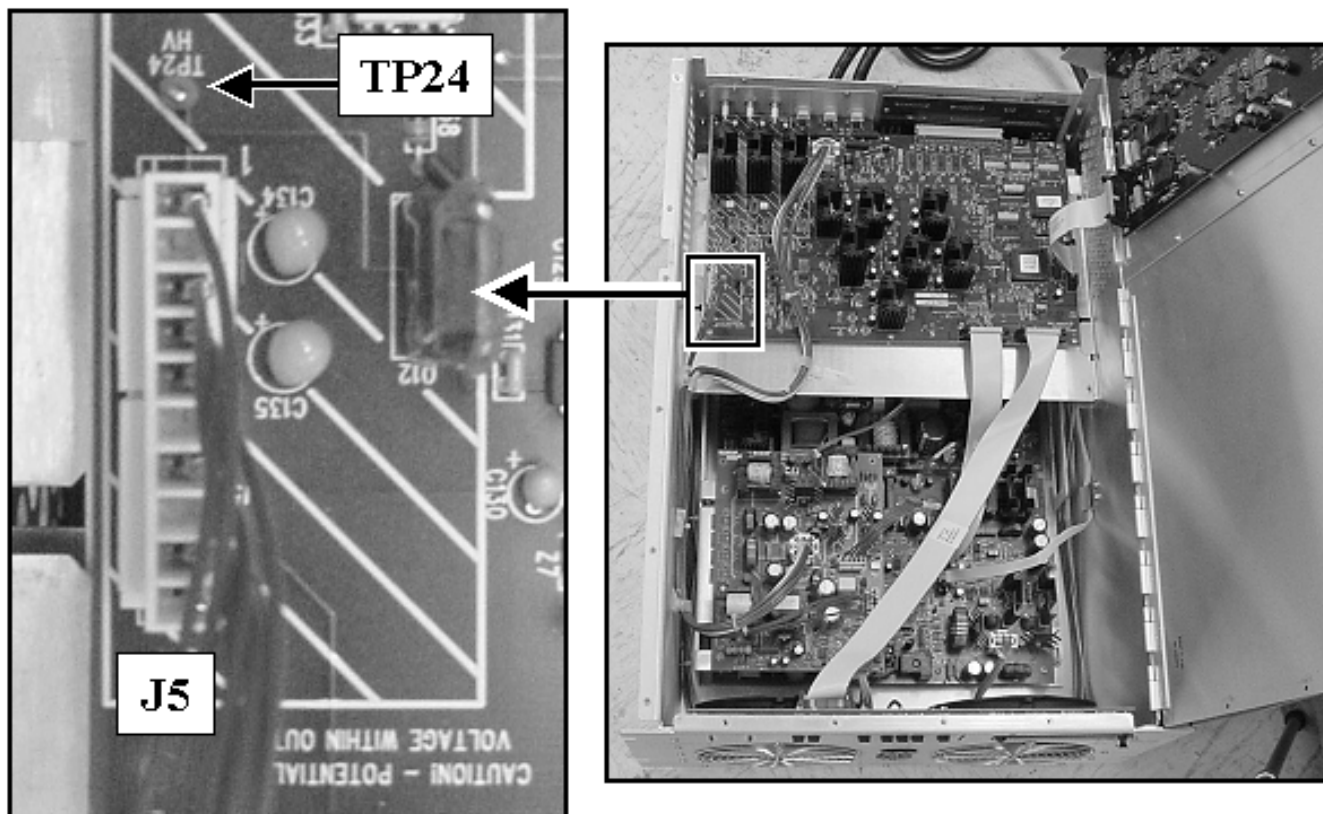
2. Wait 5 minutes after placing CB4 in the down (OFF) position for the high voltage supply in the SSM to discharge before performing the next step.
3. Refer to **APPENDIX A – REMOVING AND REPLACING SSM CHASSIS LID** and remove SSM lid.
4. Obtain a Digital Multimeter (DMM) with clip-on test leads that is capable of measuring 1000 VDC or more. Configure DMM to measure at least 1000VDC.
5. Refer to Illustration 7-2 and locate the CPD Board in the upper, rear-end of the SSM. The CPD Board is mounted on an aluminum plate above the Buck/H-Bridge Board.



LOCATION OF HV BOARD AND CPD BOARD INSIDE SSM
ILLUSTRATION 7-2

6. Refer to Illustration 7-3 and attach one DMM test lead to **TP24 (HV)** on the CPD Board and the other to the SSM chassis.

7-2 Poor Image Quality Due to Loss of High Voltage Bias (Continued)



LOCATION OF TP24 (HV) ON THE CPD BOARD
ILLUSTRATION 7-3

7. Close SSM lid and let it rest on test leads.
8. Place the **CB4 SYSTEM SUPPORT MODULE** Circuit Breaker in the lower-right corner rear of the SSM in the up (ON) position.

7-2 Poor Image Quality Due to Loss of High Voltage Bias (Continued)

9. Refer to Table 7-1 and setup for a body scan.

TABLE 7-1
SCAN PROTOCOL: BODY MODE

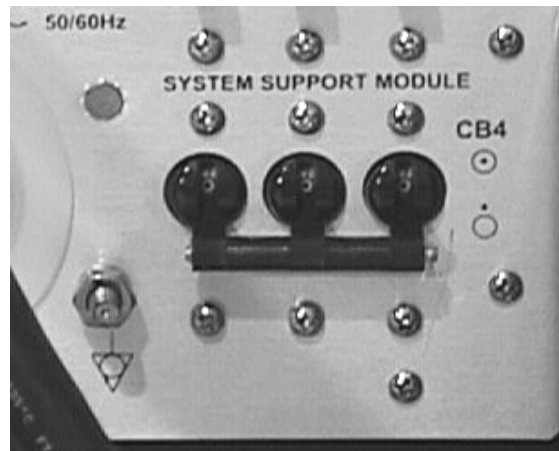
Note: This is the alternate proprietary procedure available for GE use and for sites with a valid Advanced Service Package Limited License. Any other protocol utilizing the body coil would also work for this check.

- A. **[New Pt]**
 - Id: **geservice<ENTER>**
 - Name: **Bias Check**
 - Weight (Lb.): **111<ENTER>**
 - Set Patient Protocols to **Service**.
- B. At front enclosure:
 - Landmark in the Head area—remove any coils.
 - press **LANDMARK**.
 - press **MOVE TO SCAN**.
- C. In the Patient Position Protocol field:
 - type **o.45.1<ENTER>**(o=Other, 45.1 =series) to load the body protocol
 - OR** select **other** and select protocol **45** and select series **1**.
- D. **[Save Series]**
- E. **[Manual Prescan]** (System should pulse once and then fault with a TR fault. Since the body transmit cable is disconnected from J4 Body Output on the RFI, this fault is normal.)
- F. **[Done]** (Select **Done** if for some reason system does not stop pulsing.)

10. Verify **BODY** LED is illuminated on front of RFI Module.
11. Look at the DMM display and observe the voltage reading.
- a. **If the voltage is between 950 VDC and 1050 VDC** then the HV hardware is working properly. Proceed to **Section 7-3 Restoration**.
 - b. **If the voltage is low and not between 950 VDC and 1050 VDC** then the tests in the remainder of this section will be performed to determine the extent of the failure.

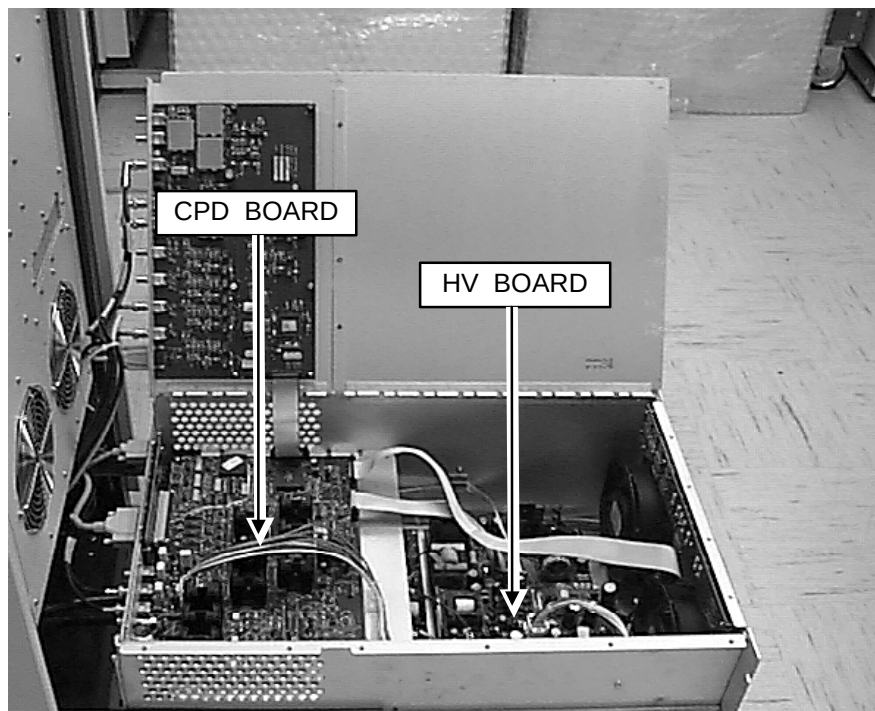
7-2 Poor Image Quality Due to Loss of High Voltage Bias (Continued)

12. Locate the **CB4 SYSTEM SUPPORT MODULE** Circuit Breaker in the lower-right corner rear of the SSM and place it in the down (OFF) position. See Illustration 7-4.



CB4 SYSTEM SUPPORT MODULE CIRCUIT BREAKER IN DOWN (OFF) POSITION
ILLUSTRATION 7-4

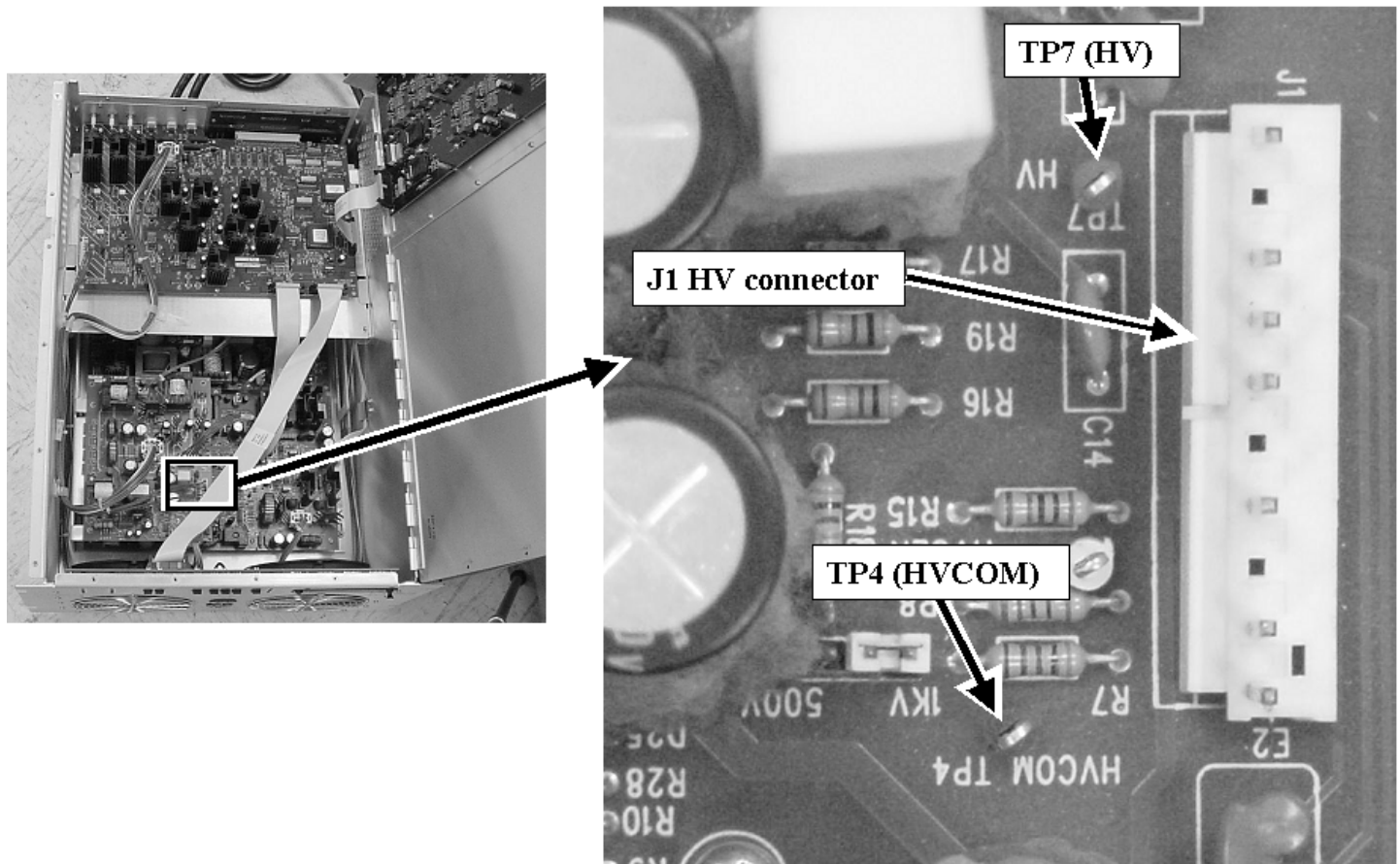
13. Wait 5 minutes after placing CB4 in the down (OFF) position for the high voltage supply in the SSM to discharge before performing the next step.
14. Open the SSM lid.
15. Refer to Illustration 7-5 and locate the HV Board in the upper, front-end of the SSM. The HV Board is mounted on stand-offs above the Dock/Light Board.



LOCATION OF HV BOARD AND CPD BOARD INSIDE SSM
ILLUSTRATION 7-5

7-2 Poor Image Quality Due to Loss of High Voltage Bias (Continued)

16. Refer to Illustration 7-6 and locate test points **TP7 (HV)** and **TP4 (HVCOM)** on the HV Board.

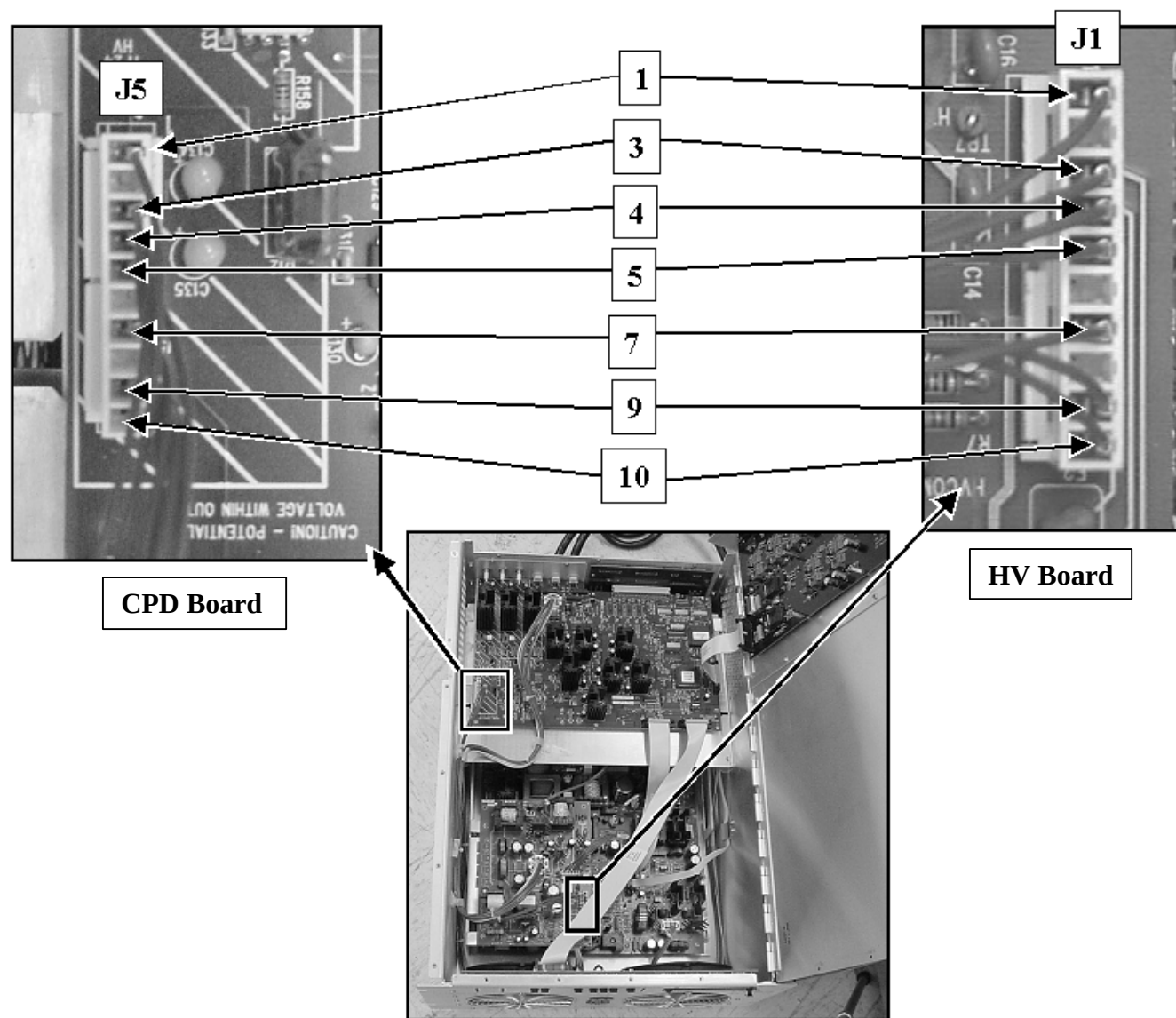


LOCATION OF J1, TP7, AND TP4 ON THE HV BOARD
ILLUSTRATION 7-6

17. Configure the DMM to measure 1000VDC and then carefully connect the positive DMM lead to **TP7 (HV)** and negative DMM lead to **TP4 (HVCOM)**. Avoid allowing the bare test leads to come into contact with fingers.
18. Verify that the DMM shows less than 10VDC between these two test points. If necessary, wait for the voltage to drop below 10VDC before proceeding.

7-2 Poor Image Quality Due to Loss of High Voltage Bias (Continued)

19. Refer to Illustration 7-7 and locate the red HV Power Cable (Erbtec part #550010) interconnecting J5 on the CPD Board to J1 on the HV Board.



HV POWER CABLE CONNECTIONS AND PINS
ILLUSTRATION 7-7

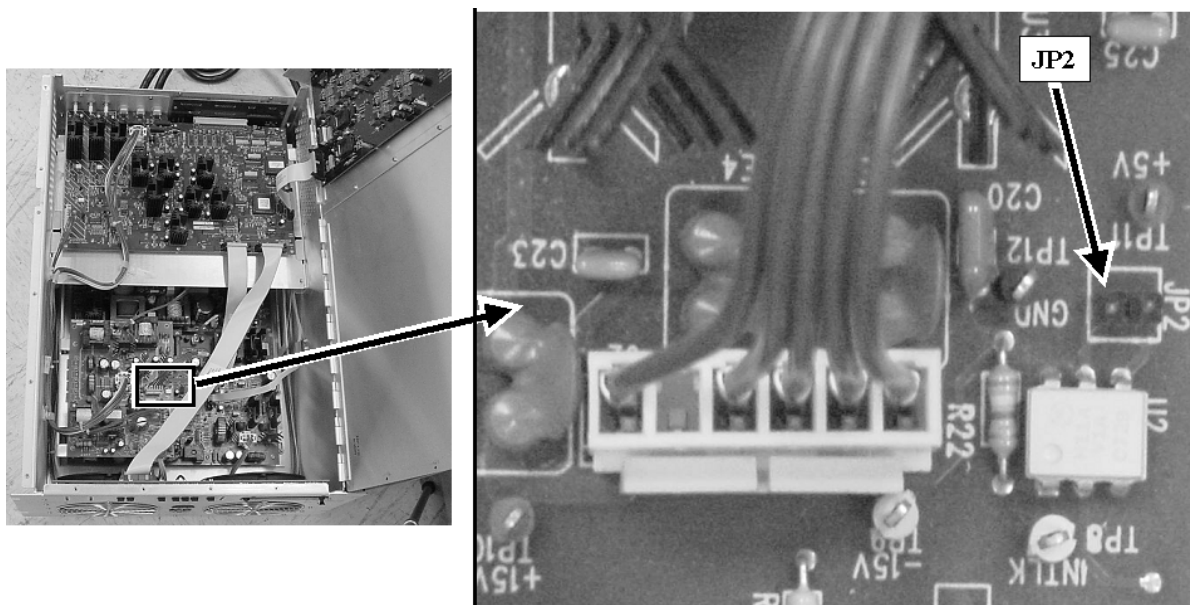
7-2 Poor Image Quality Due to Loss of High Voltage Bias (Continued)

20. Refer to Table 7-2 and, with the HV Power Cable connected as shown in Illustration 7-7, use the DMM to perform a continuity check (measure the resistance) between each of the pins of CPD Board J5 and HV Board J1 and confirm that a good connection (near zero resistance) exists between the corresponding pins of each of the cable connectors.

TABLE 7-2
J5 CPD BOARD AND J1 HV BOARD CABLE CONTINUITY CHECKS

Measure Between
J1-1 and J5-1
J1-3 and J5-3
J1-4 and J5-4
J1-5 and J5-5
J1-7 and J5-7
J1-9 and J5-9
J1-10 and J5-10

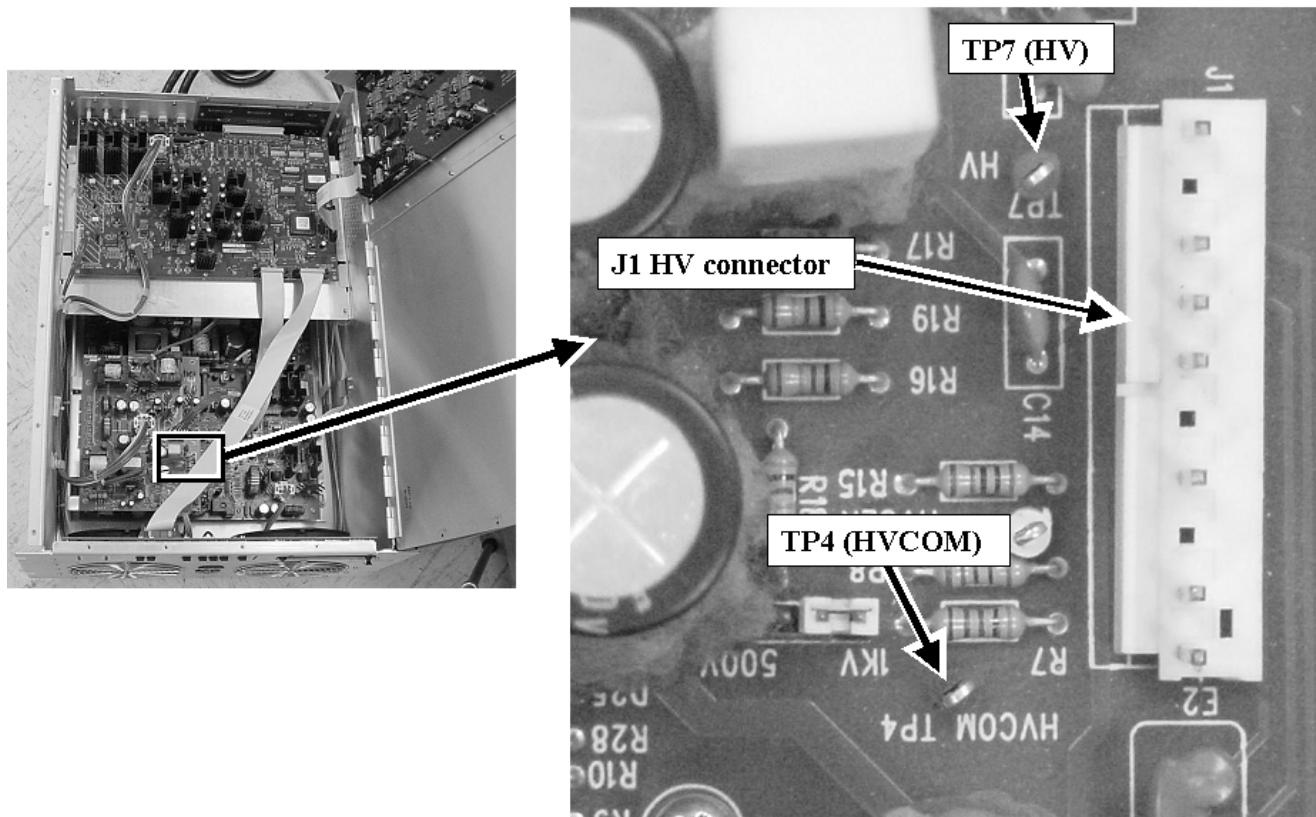
21. Replace or repair the red HV Power Cable if a high resistance is measured between any of the corresponding connector pins referenced in Table 7-2.
22. Refer to Illustration 7-8. Locate the JP2 jumper pins on the HV Board (High Voltage Board) and short these together with a spare jumper or piece of wire.



LOCATION OF JUMPER JP2 ON THE HV BOARD
ILLUSTRATION 7-8

7-2 Poor Image Quality Due to Loss of High Voltage Bias (Continued)

23. Refer to Illustration 7-9 and disconnect the red HV Power Cable at J1 on the HV Board and set it aside.



LOCATION OF J1, TP7, AND TP4 ON THE HV BOARD
ILLUSTRATION 7-9

24. Configure the DMM to measure 1000VDC and then connect the positive DMM lead to **TP7 (HV)** and negative DMM lead to **TP4 (HVCOM)**.
25. Close SSM lid and let it rest on test leads.
26. Place the **CB4 SYSTEM SUPPORT MODULE** Circuit Breaker in the lower-right corner rear of the SSM in the up (ON) position.
27. Verify that the reading on the DMM display is between 950 VDC and 1050 VDC.
- If the voltage is low and is not between 950 VDC and 1050 VDC** then the HV Board has failed. The CPD Board may have shorted and caused it to fail. Replace both the CPD Board and the HV Board.
 - If the voltage is between 950 VDC and 1050 VDC** then the CPD Board failed but did not damage the HV Board. Replace the CPD Board.
28. Locate the **CB4 SYSTEM SUPPORT MODULE** Circuit Breaker in the lower-right corner rear of the SSM and place it in the down (OFF) position.
29. Watch the DMM display and wait for the high voltage to discharge down below 10 volts.

7-2 Poor Image Quality Due to Loss of High Voltage Bias (Continued)

30. Reconnect the HV Power Cable to J1 on the HV Board.
31. Remove the jumper or shorting wire from between the pins of JP2 on the HV Board.
32. Remove the DMM test leads from **TP7** and **TP4** on the HV Board and close the SSM lid cover.
33. Proceed to **Section 7-3 Restoration**.

7-3 RESTORATION

1. Confirm that the black fault switches on the front of the SSM are in the up (ENable) position.
2. Perform actions in **Section A-2 Replacing SSM Chassis Lid**.
3. Place the **CB4 SYSTEM SUPPORT MODULE** Circuit Breaker in the lower-right corner rear of the SSM in the up (ON) position.
4. Reset the TPS from the Service Desktop Manager Window.
5. Perform one head and one body scan.
6. Replace the front cover on the RF Cabinet.

8 - TROUBLESHOOTING TR DRIVER FAULTS

8-1 General Theory Overview

The Driver Module now provides all head, body, and multicoil and TR biasing functions as well as all dynamic disable and direct drive biasing.

APPENDIX A – REMOVING AND REPLACING SSM CHASSIS LID

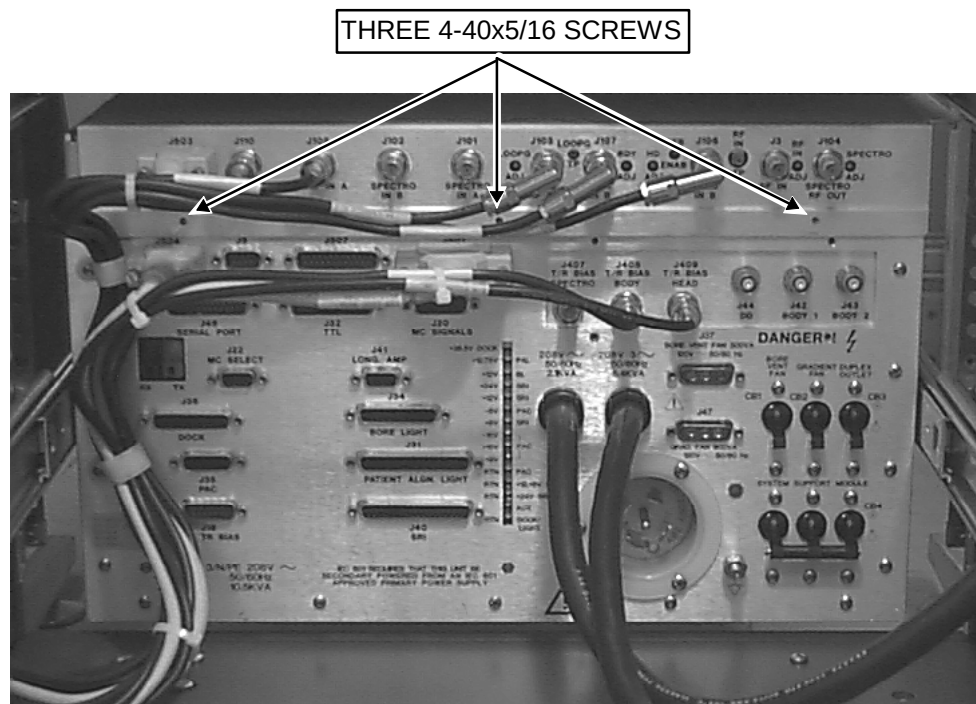
The process works the same regardless of whether the SSM is inside an RF/PDU or SRF Cabinet.

A-1 Removing SSM Chassis Lid



POSSIBLE PERSONAL INJURY! AVOID SERIOUS INJURY OR DEATH BY ELECTROCUTION. MAKE CERTAIN THAT SYSTEM SUPPORT MODULE CB4 CIRCUIT BREAKER ON THE REAR OF THE SSM IS IN THE DOWN (OFF) POSITION. BE AWARE THAT THIS WILL REMOVE POWER FROM HIGH VOLTAGE DYNAMIC DISABLE AND INDUCTIVE DRIVE DISABLE CIRCUITS AND THE LOW-VOLTAGE TR DRIVER CIRCUITS BUT DOES NOT REMOVE AC POWER FROM THE NB AMP RECEPTACLE, SPECTRO AMP RECEPTACLE, FRONT SERVICE OUTLETS, OR GRADIENT BLOWER FAN.

1. Confirm that System Support Module CB4 circuit breaker on the rear of the SSM is in the down (OFF) position.
2. Remove the three 4-40x5/16 panhead screws that secure the APM lid to the rear panel (see Illustration A-1). A few cables may need to be disconnected first.

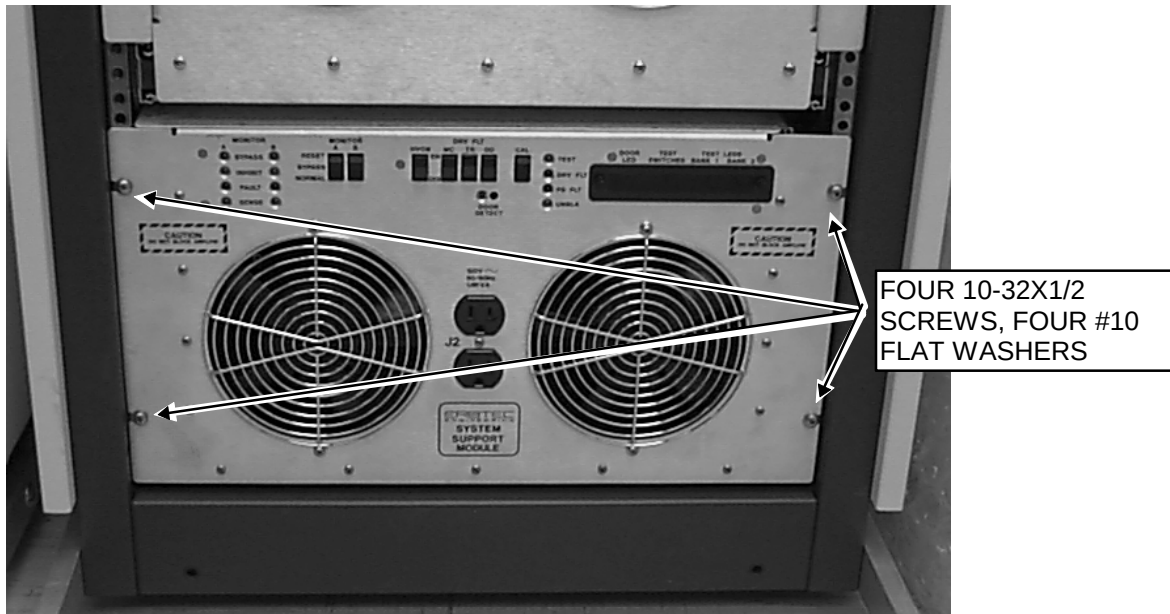


REMOVAL OF ANALOG PROCESSOR MODULE LID FROM REAR PANEL
ILLUSTRATION A-1

3. Remove the front door from the cabinet.

A-1 Removing SSM Chassis Lid (Continued)

4. Remove the four 10-32x1/2 screws and four #10 flat washers that secure the SSM chassis to the cabinet (see Illustration A-2).



REMOVAL OF SYSTEM SUPPORT MODULE
ILLUSTRATION A-2

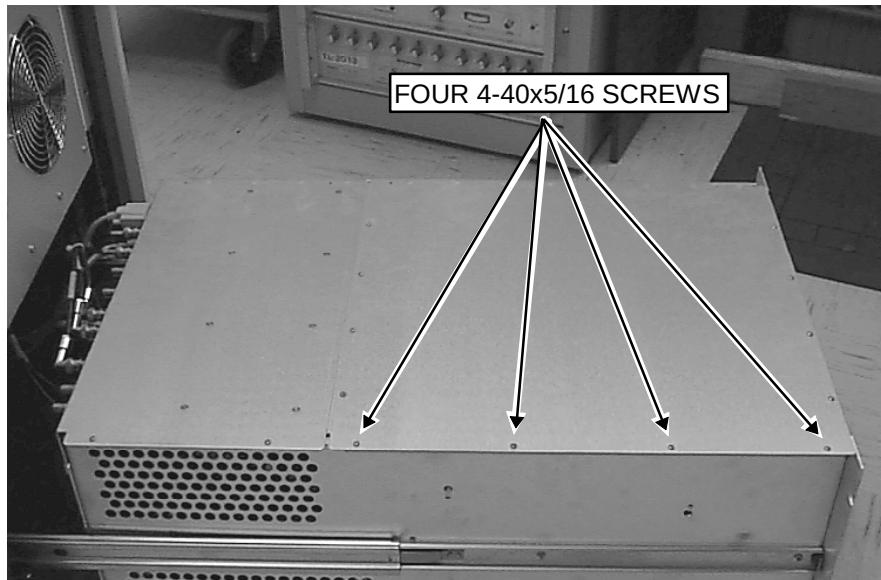
5. Remove all external cabling from the rear of the SSM.
6. Slide out the SSM from the cabinet.
7. Remove the four 4-40x5/16 panhead screws that secure the front lid to the front panel (see Illustration A-3).



REMOVAL OF SYSTEM SUPPORT MODULE LID FROM FRONT PANEL
ILLUSTRATION A-3

A-1 Removing SSM Chassis Lid (Continued)

8. Remove the four 4-40x5/16 panhead screws that secure the front lid to the chassis (see Illustration A-4).



REMOVAL OF SYSTEM SUPPORT MODULE LID FROM CHASSIS
ILLUSTRATION A-4

9. Remove the two 4-40x5/16 panhead screws that secure the APM lid to the chassis (see Illustration A-5).



REMOVAL OF ANALOG PROCESSOR MODULE LID FROM CHASSIS
ILLUSTRATION A-5

10. Grasp the edge of the module lid where the APM lid meets the front lid, and lift the lid to a vertical position. Use care to ensure that the APM cables do not catch on the bottom of the RFI and that there is sufficient slack.
11. Return to the previous procedure.

A-2 Replacing SSM Chassis Lid

1. Close the front lid. Install eight 4-40 x 5/16 panhead screws: four to the APM lid, and four to the front panel.
2. Attach six 4-40x5/16 panhead screws that secure both lids to chassis. Tighten all screws in steps 4, 5, and 6.
3. Slide the module back into the cabinet. (Press the slide locks to enable the module to slide completely into the cabinet.)
4. Secure SSM to cabinet with four 10-32x1/2 panhead screws and four #10 flat washers.
5. Install three 4-40x5/16 panhead screws that secure the APM lid to the rear panel.
6. Return to previous procedure.

REVISION HISTORY

REV	DATE	AUTHOR	PRIMARY REASONS FOR CHANGE
1	March 1, 2008	Don Thome	Initial Release